

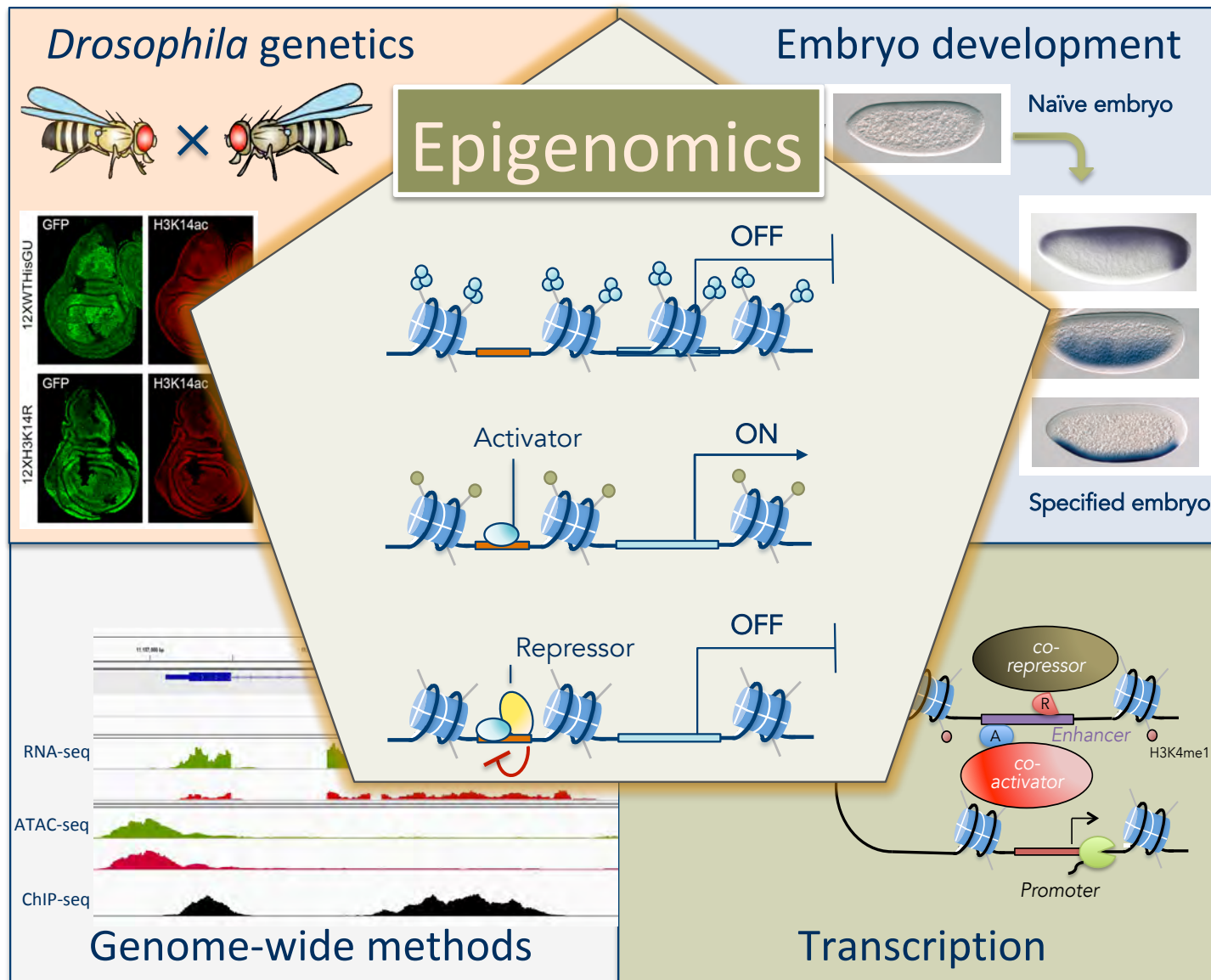
Pioneering chromatin landscapes in embryo development – epigenomic methods reveal how gene expression patterns are formed and maintained

Mattias Mannervik

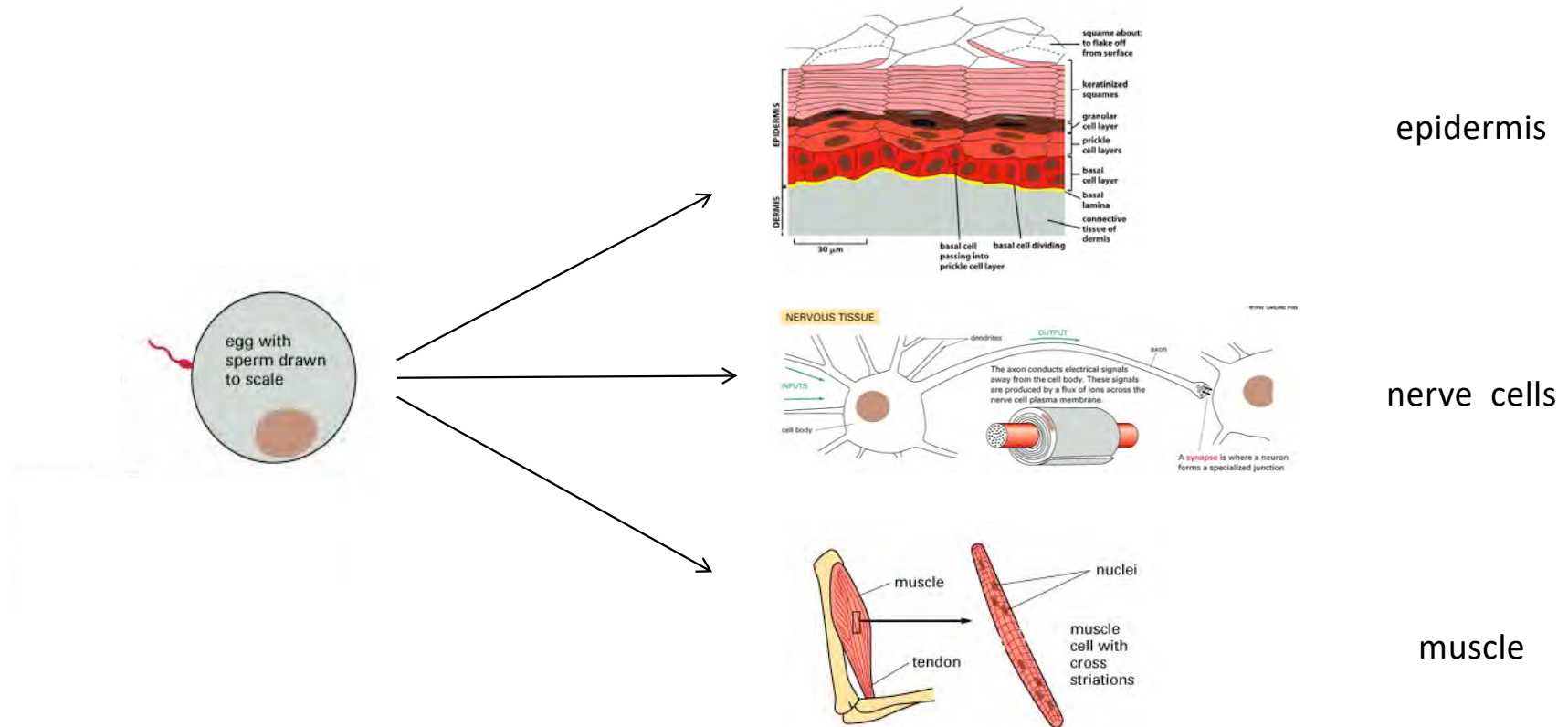
Dept. Molecular Biosciences, The Wenner-Gren Institute, Stockholm University



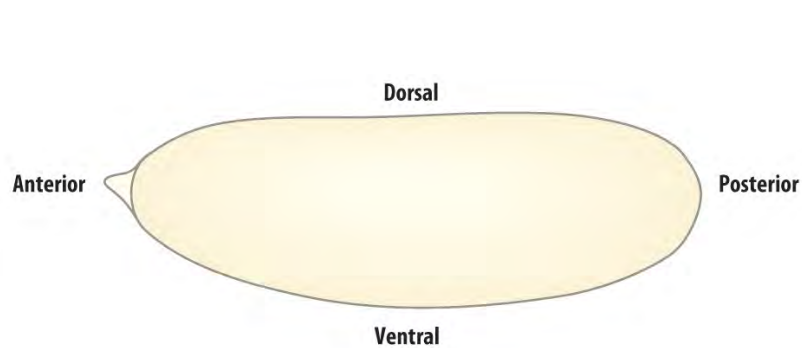
Stockholms
universitet



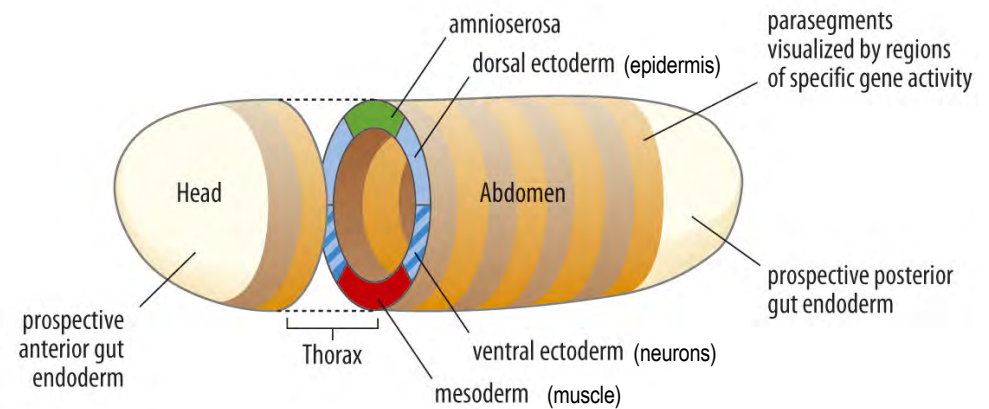
How does a single cell generate an organism with hundreds of different cell types?



Expression of different genes in distinct patterns in the embryo

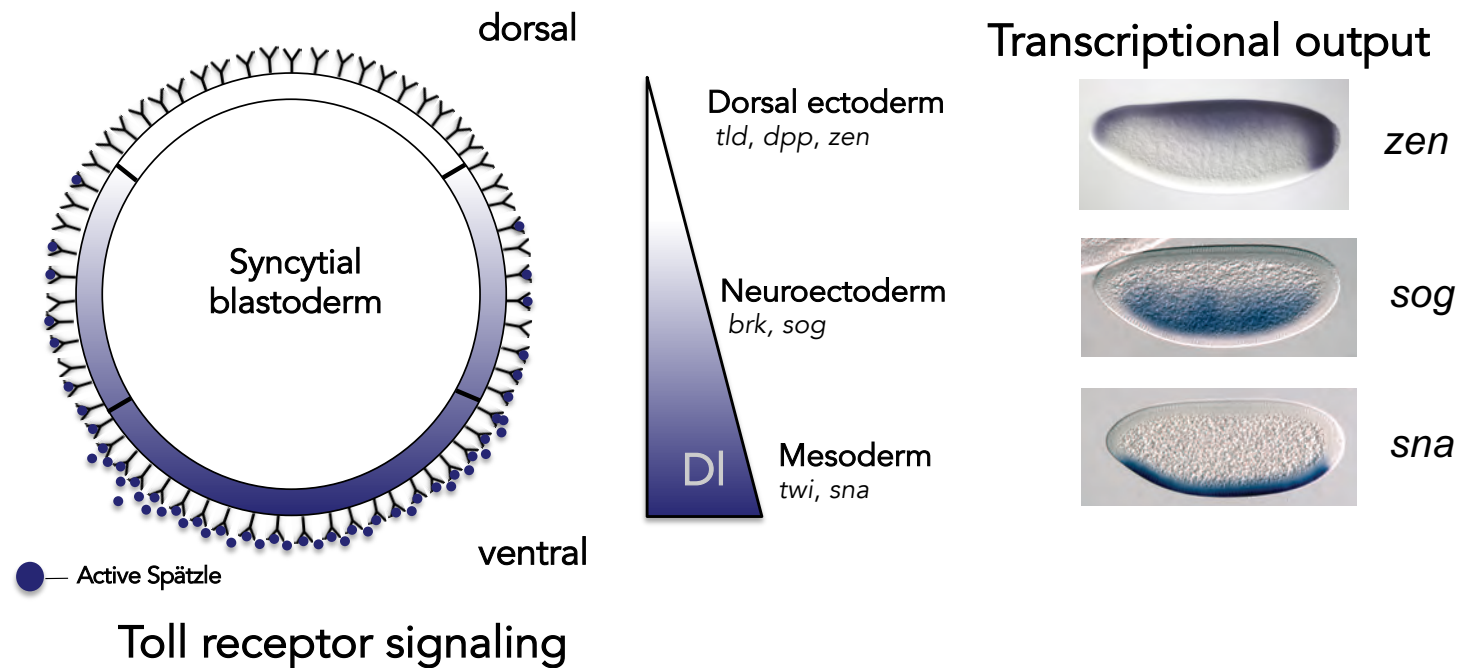


Fertilized egg



Early embryo

Presence of transcription factors in different cell types



- The Toll ligand Spätzle is activated ventrally
- Leads to intranuclear concentration gradient of Dorsal protein that differentially regulates >50 target genes across the dorsal-ventral axis

Complication: DNA is not always accessible for transcription factor binding

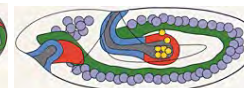
Naïve embryo



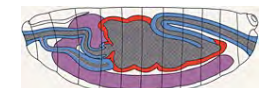
Specified



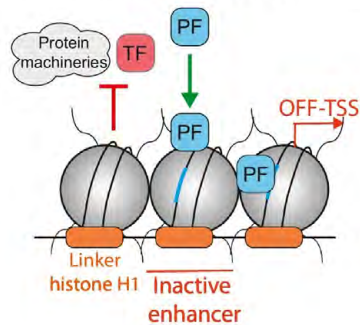
Determined



Differentiated



Naive chromatin



Complication: DNA is not always accessible for transcription factor binding

Naïve embryo



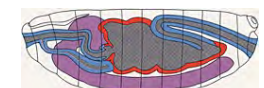
Specified



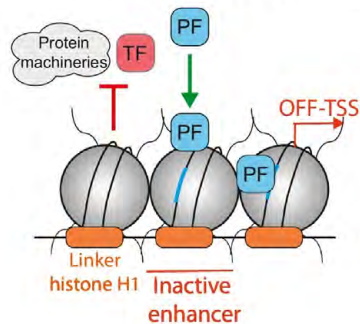
Determined



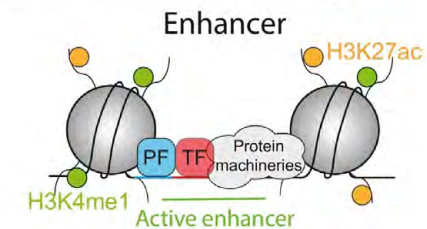
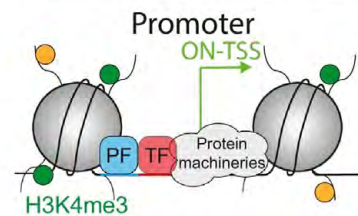
Differentiated



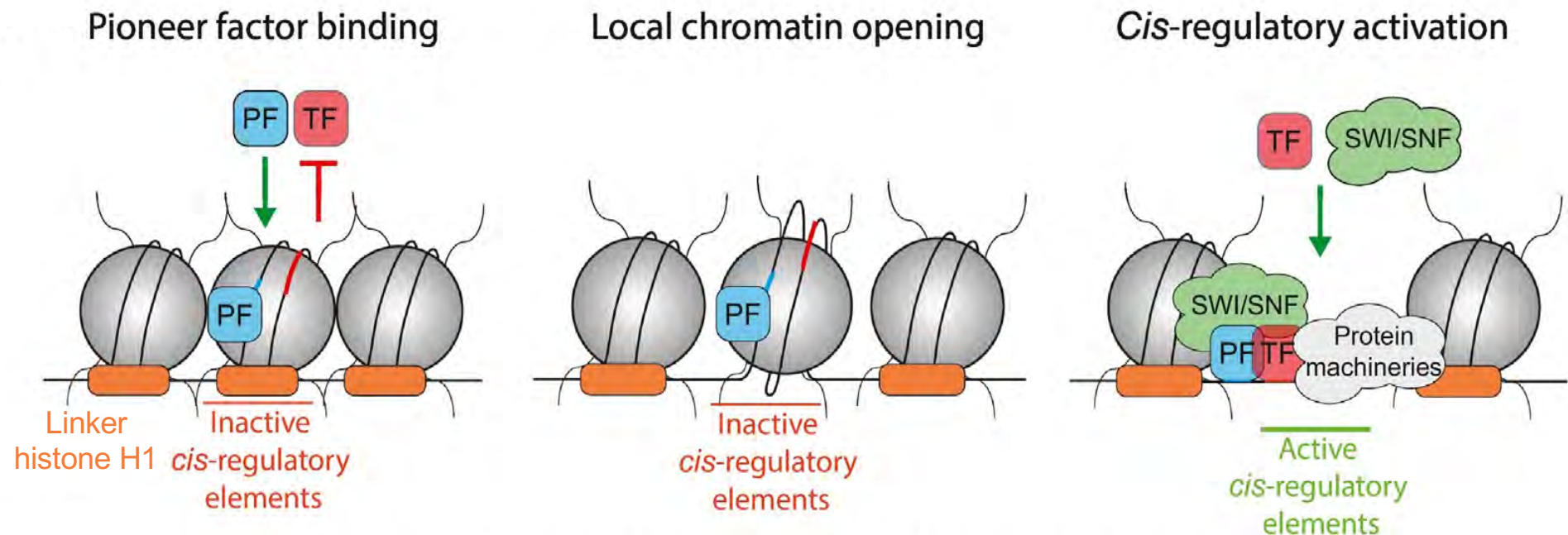
Naïve chromatin



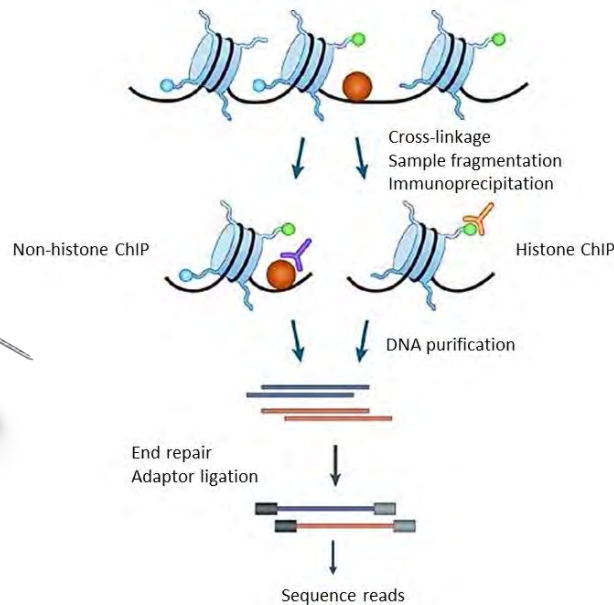
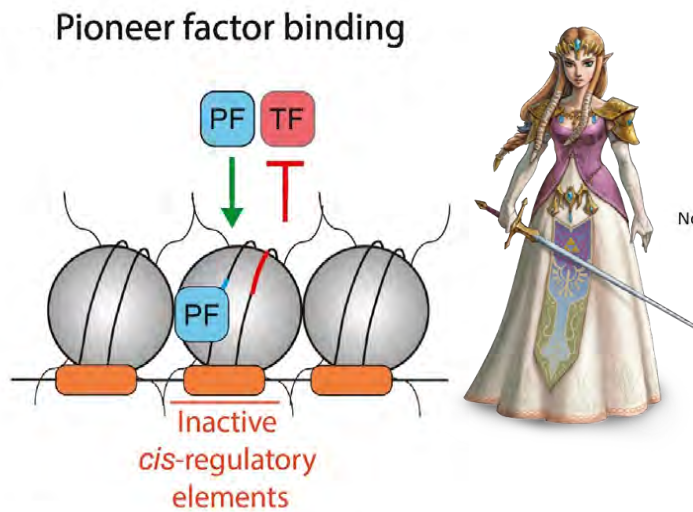
Cis-regulatory elements in euchromatin regions



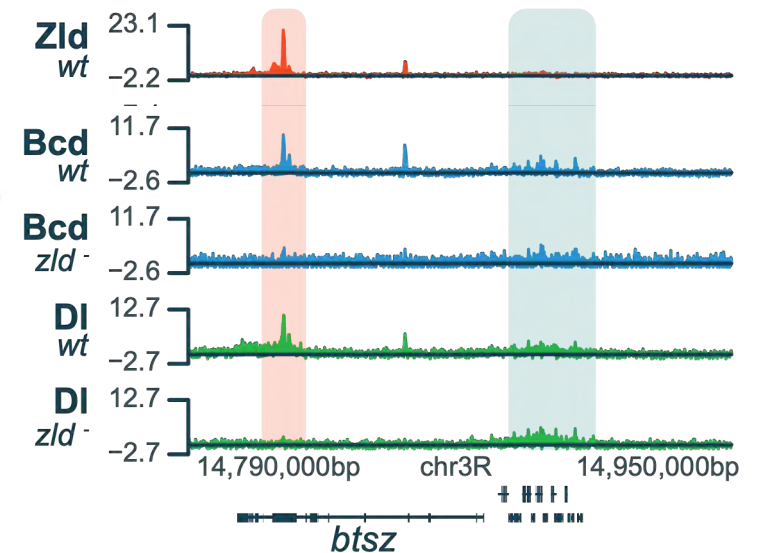
Pioneer transcription factors can access inactive DNA and facilitate binding of other proteins



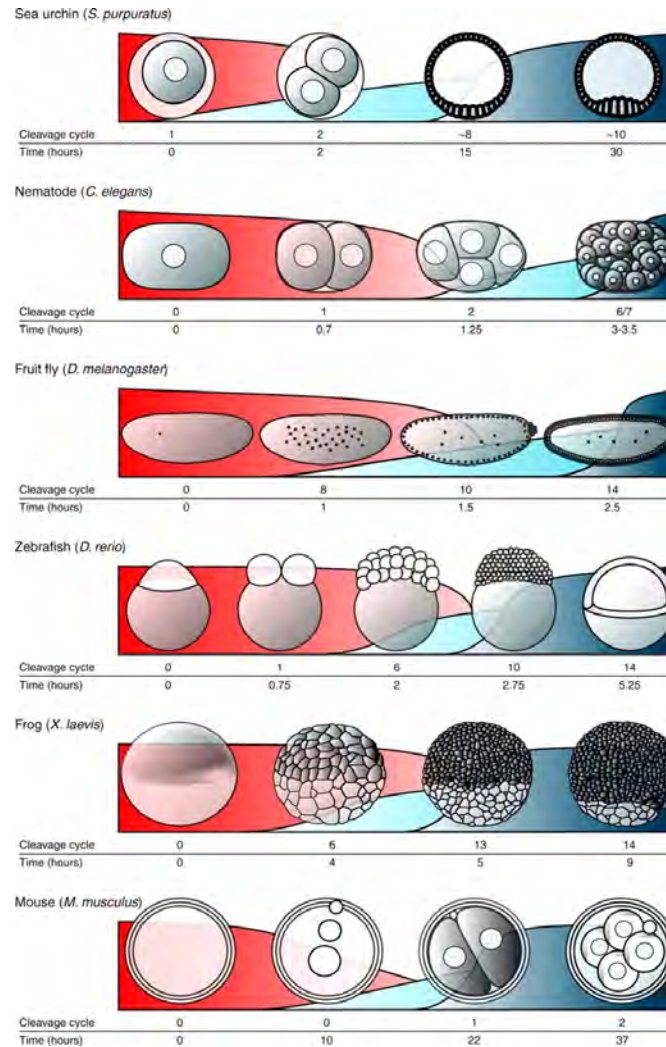
The pioneer factor Zelda (Zld) is key to zygotic genome activation in *Drosophila* embryos



ChIP-seq

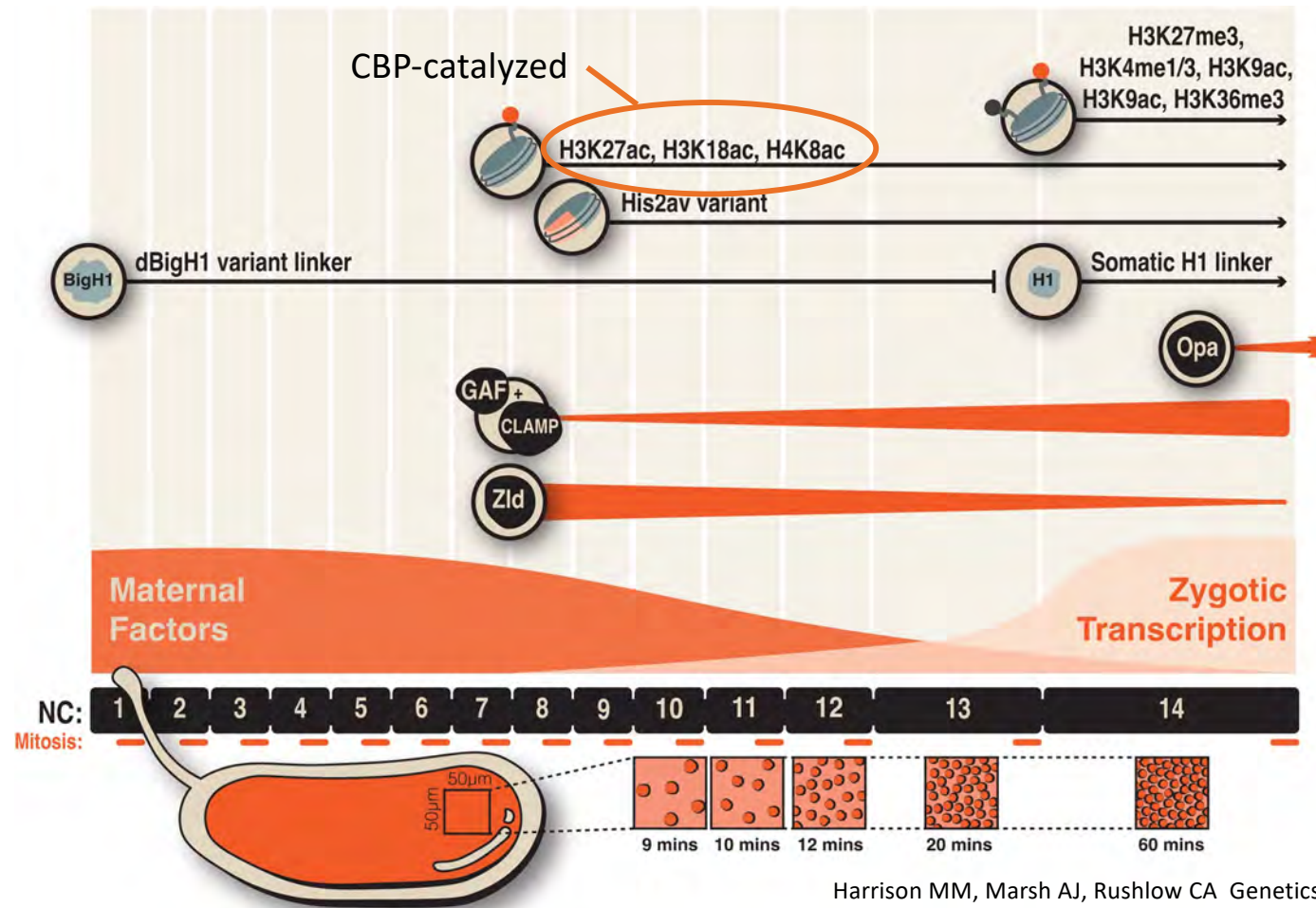


Zygotic genome activation is a conserved process in animal embryos



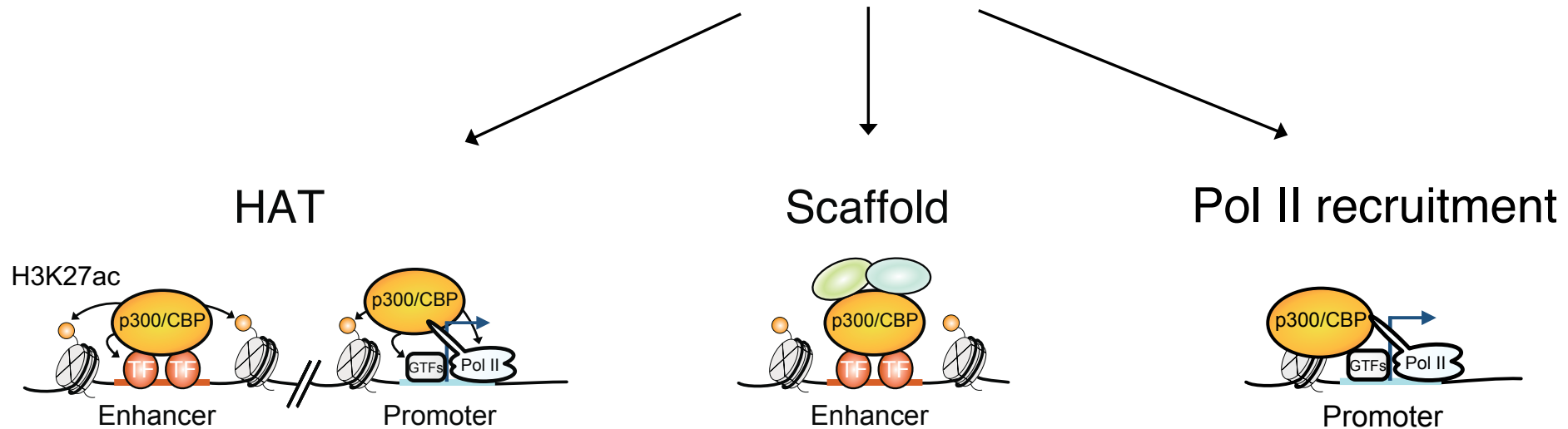
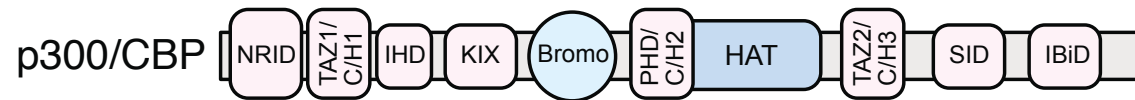
Wael Tadros, Howard D. Lipshitz, Development (2009)

CBP-catalyzed histone modifications coincide with zygotic genome activation



Harrison MM, Marsh AJ, Rushlow CA Genetics (2023) 225:iyad142

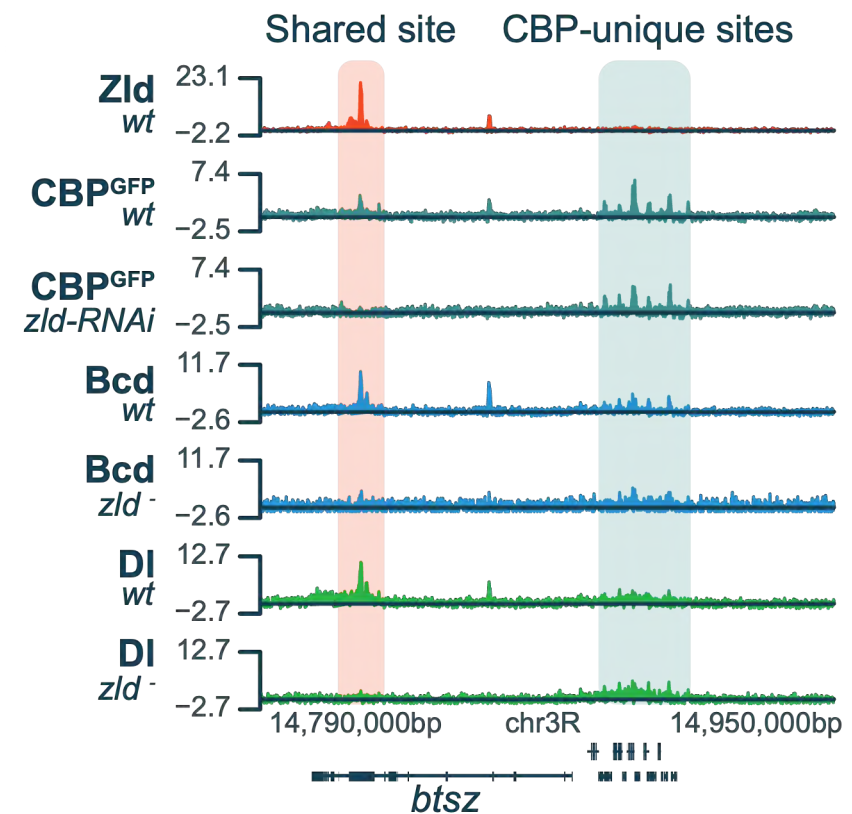
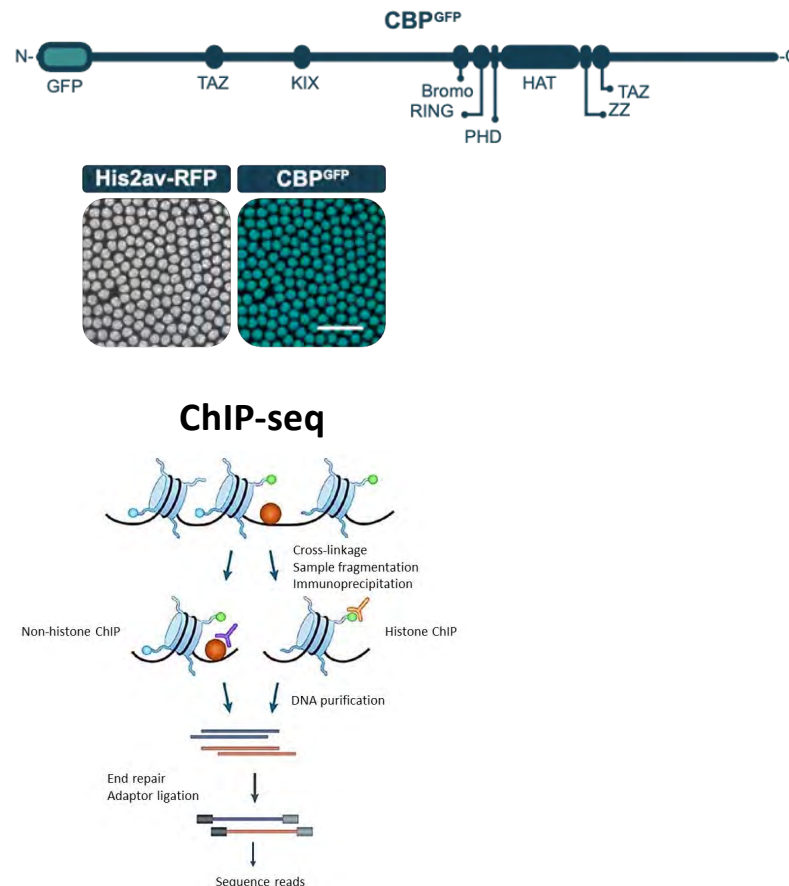
p300/CBP is a transcriptional coactivator and histone acetyltransferase (HAT)



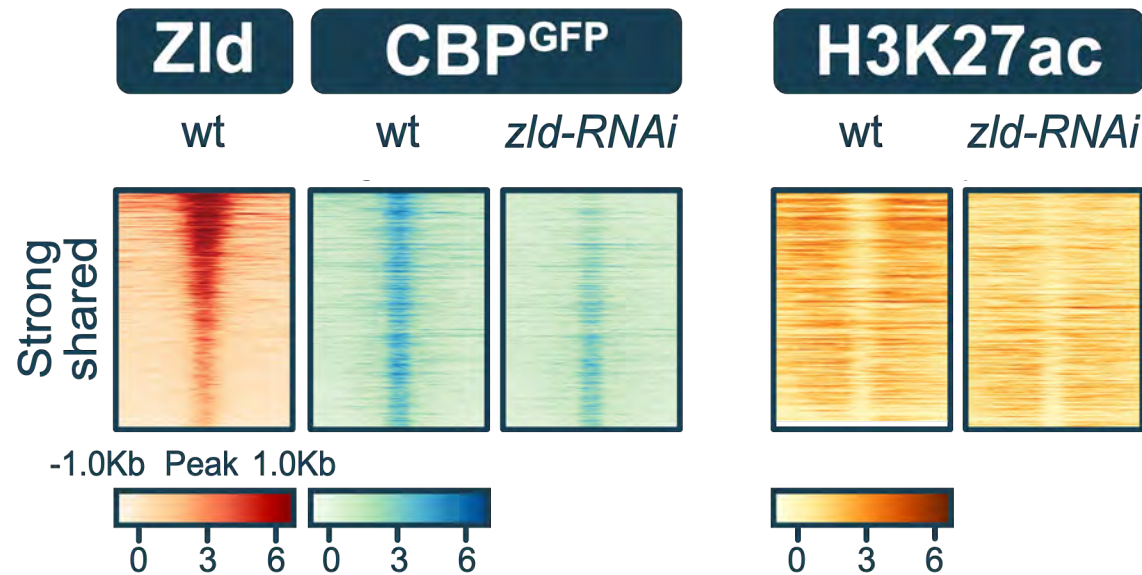
- CREB-binding protein (CBP) and its paralog p300 are transcriptional coactivators with histone acetyltransferase (HAT) enzymatic activities but lack intrinsic DNA-binding capabilities
- p300/CBP are central hubs of transcriptional networks through protein-protein interactions with many transcription factor (TF) families

Is Zld needed for CBP-mediated histone acetylation?

Zld is required for CBP binding and histone acetylation at a subset of sites

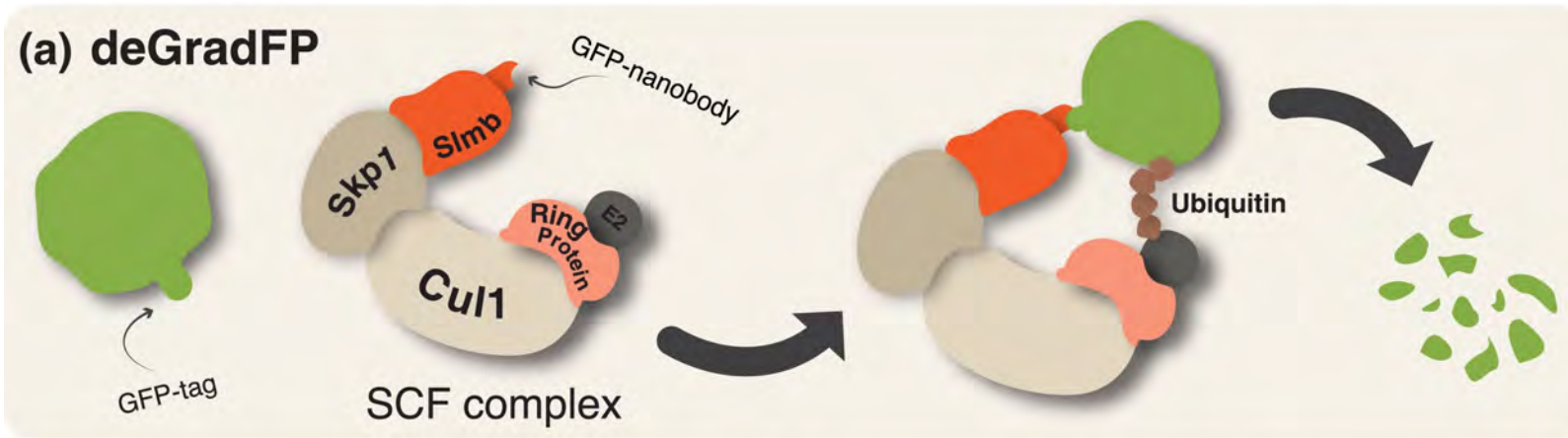
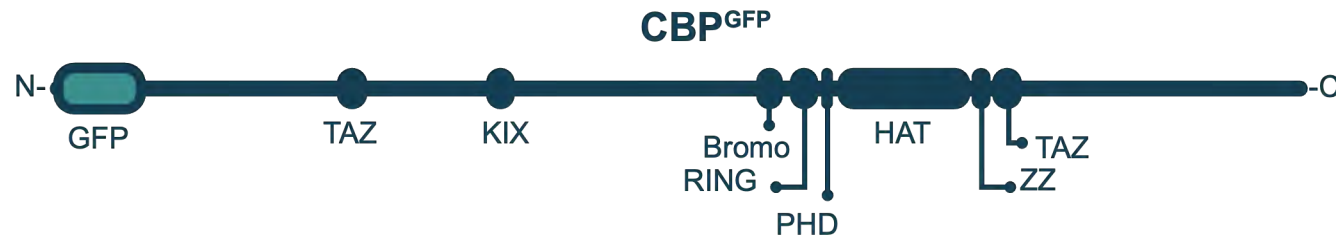


Zld is required for CBP binding and histone acetylation at a subset of sites

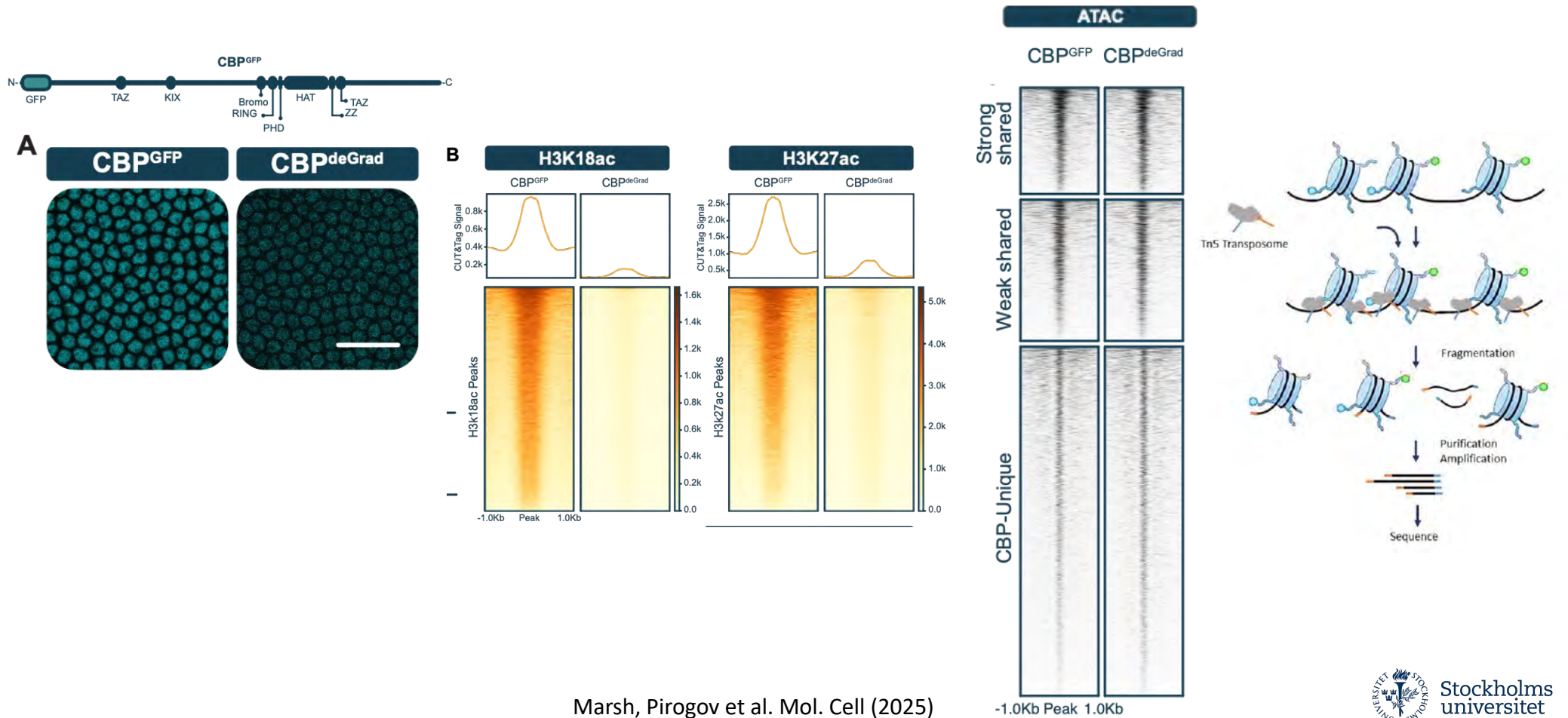


Is CBP needed for Zld pioneer activity?

Degradation of GFP-tagged CBP

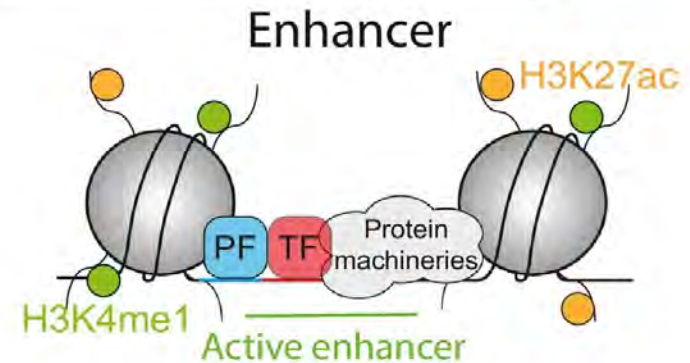
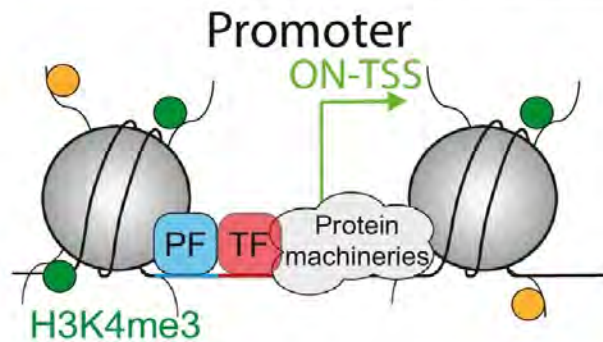


CBP and H3K27ac is not needed to form open chromatin

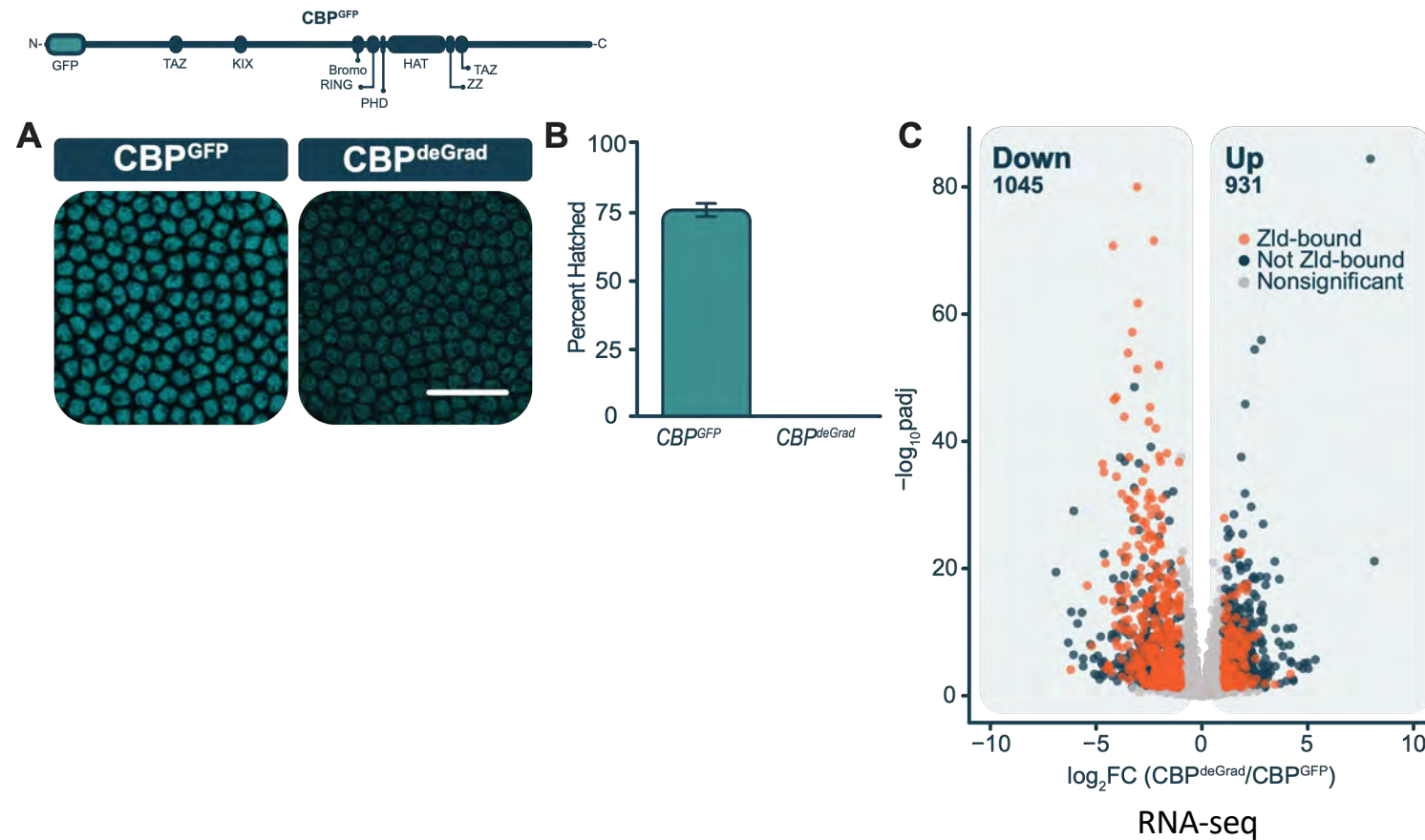


H3K27 acetylation is not needed for pioneer activity

Cis-regulatory elements in euchromatin



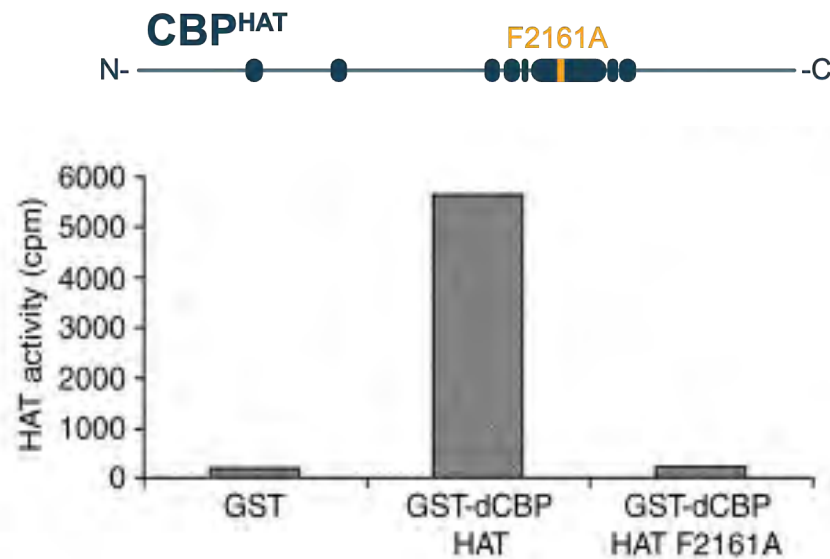
CBP is required for zygotic genome activation



Marsh, Pirogov et al. Mol. Cell (2025)

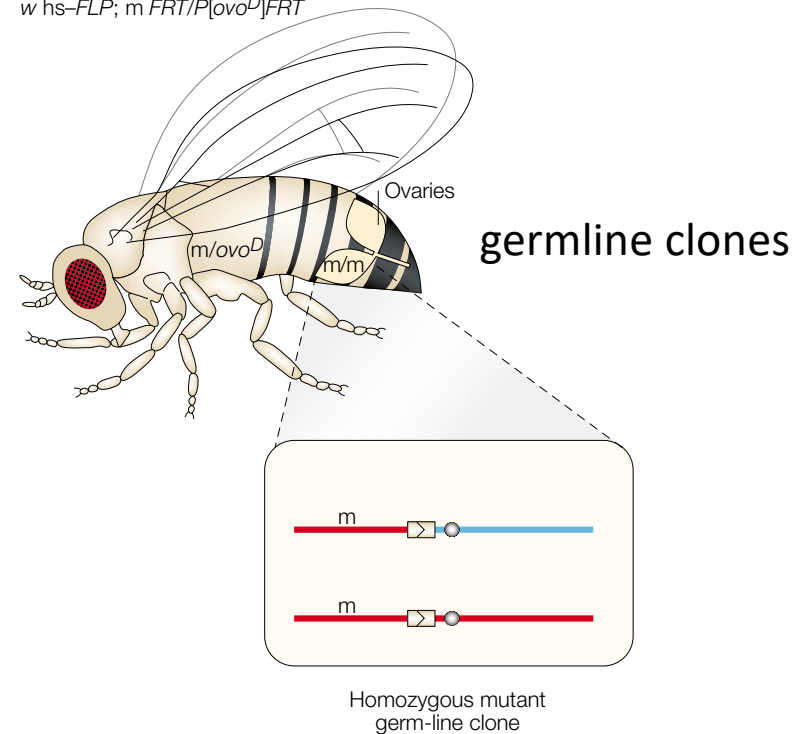
Is CBP catalytic activity needed for zygotic genome activation?

A point mutant that disrupts catalytic activity



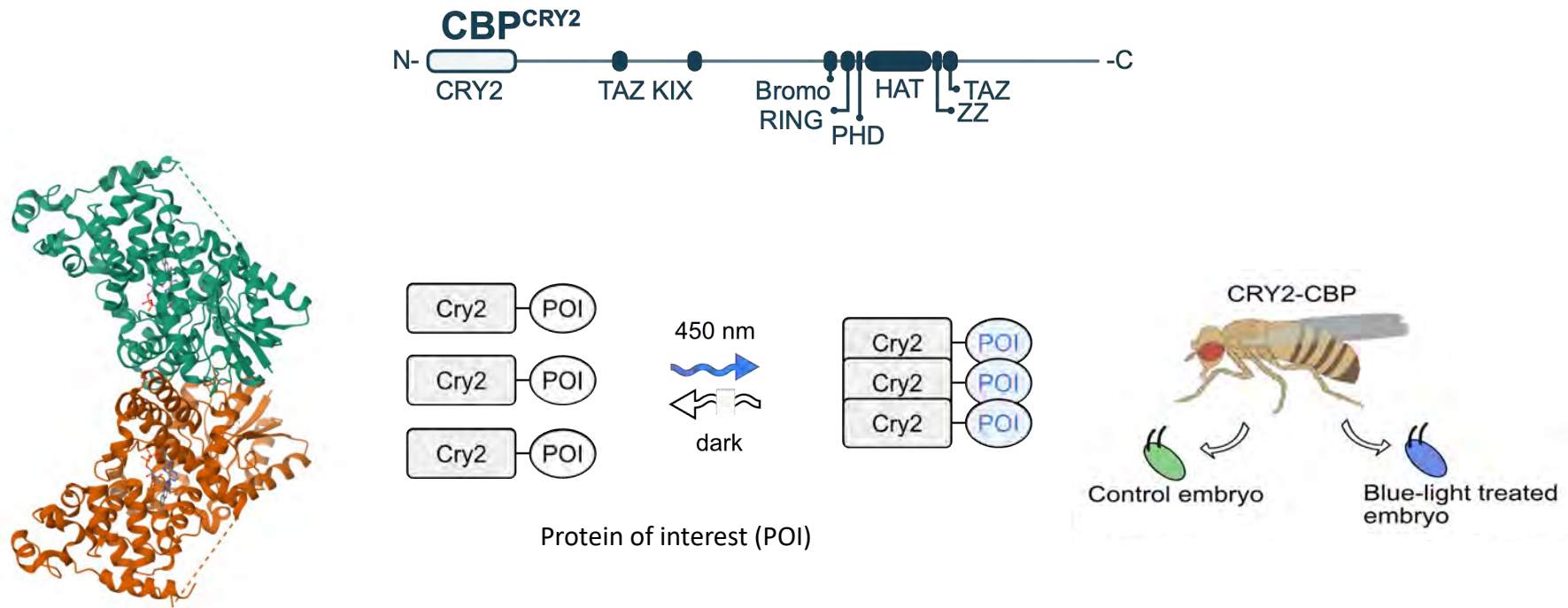
Lilja et al. Dev. Biol (2007) 305:650

w *hs-FLP*; *m FRT/P[ovo^D]**FRT*

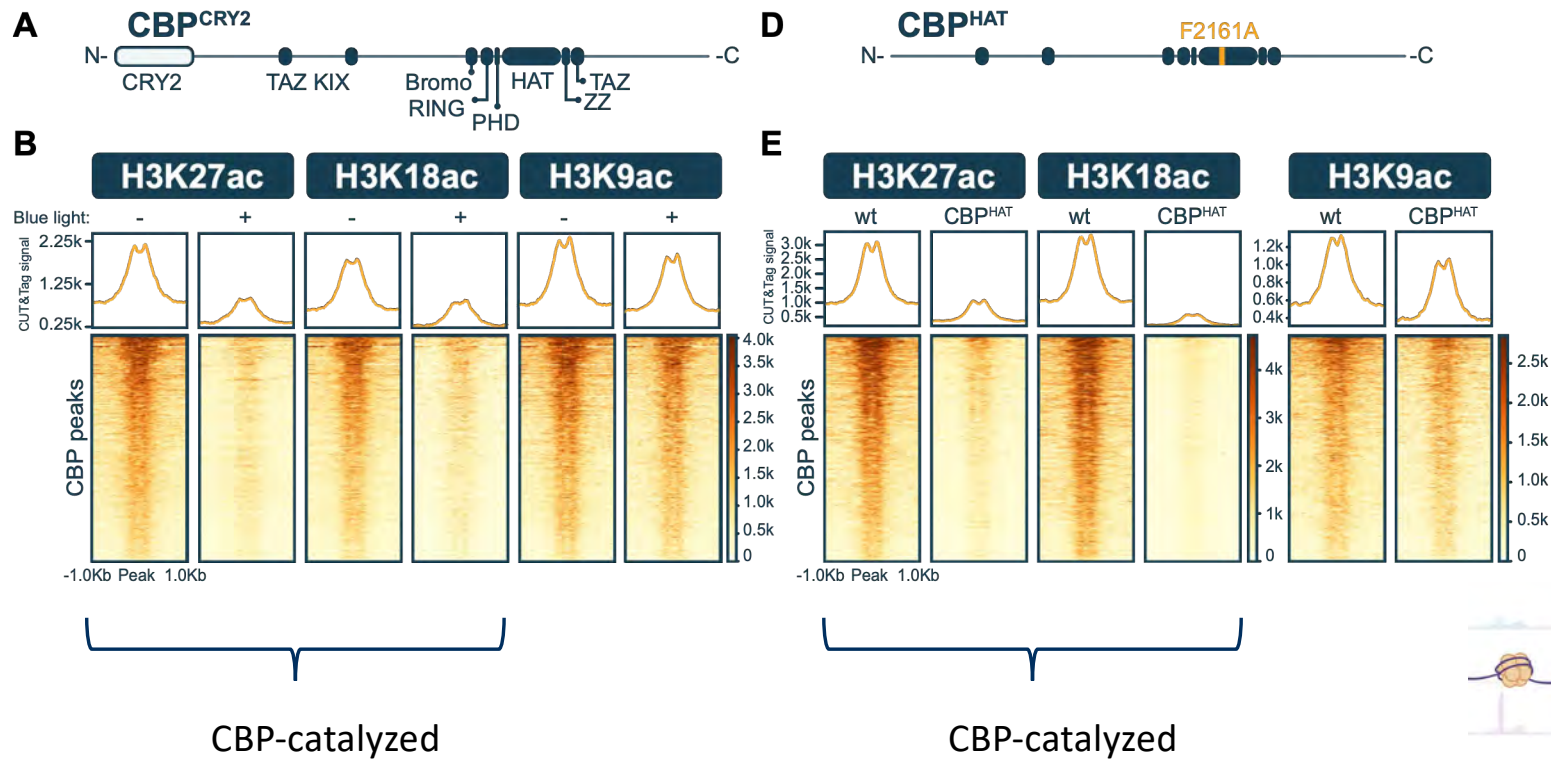


St Johnston Nature Reviews Genetics (2002) 3:176

CRY2-CBP can be inactivated by blue light (optogenetics)



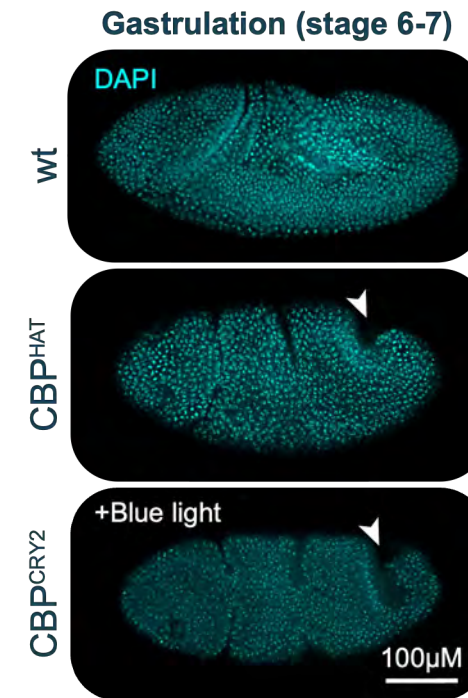
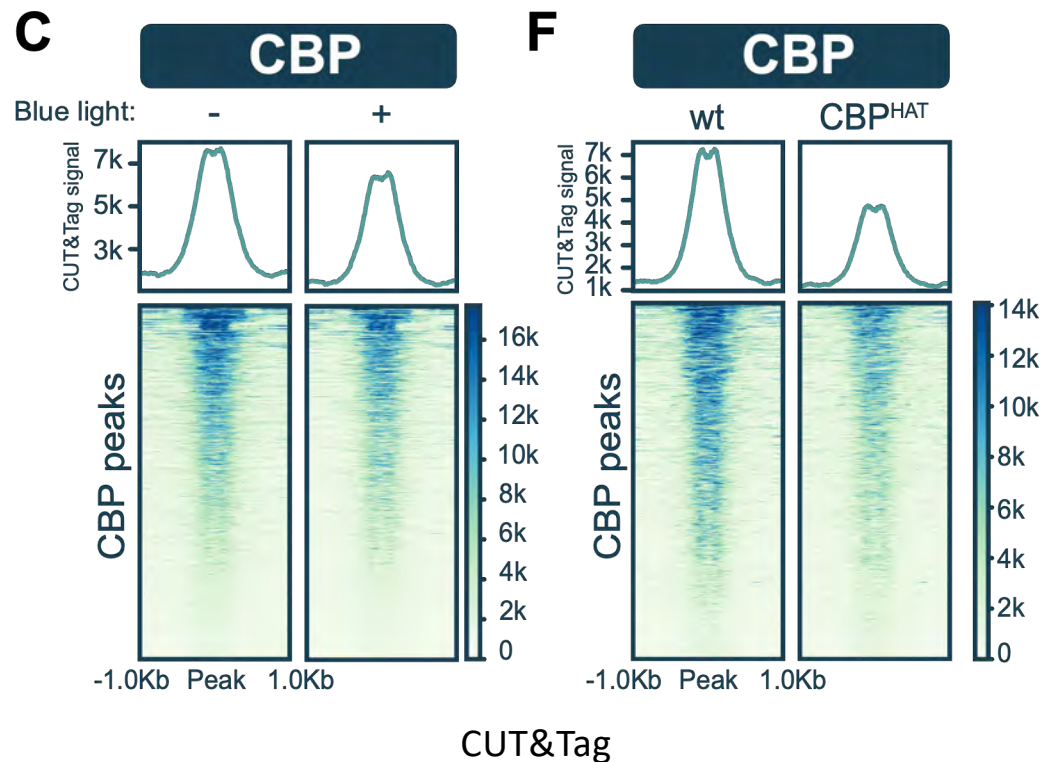
Both optogenetic inactivation and the CBP^{HAT} mutant disrupt catalytic activity



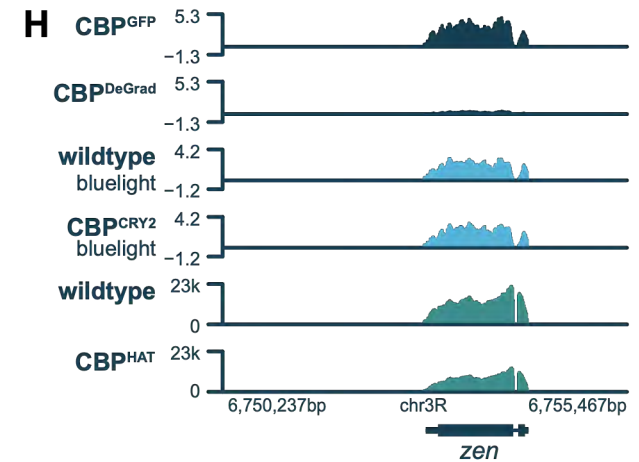
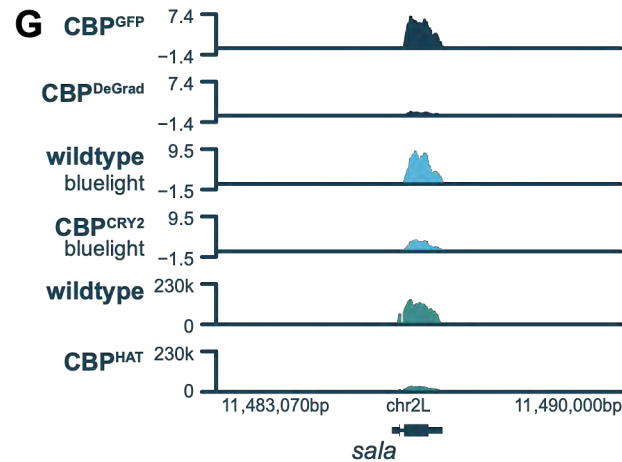
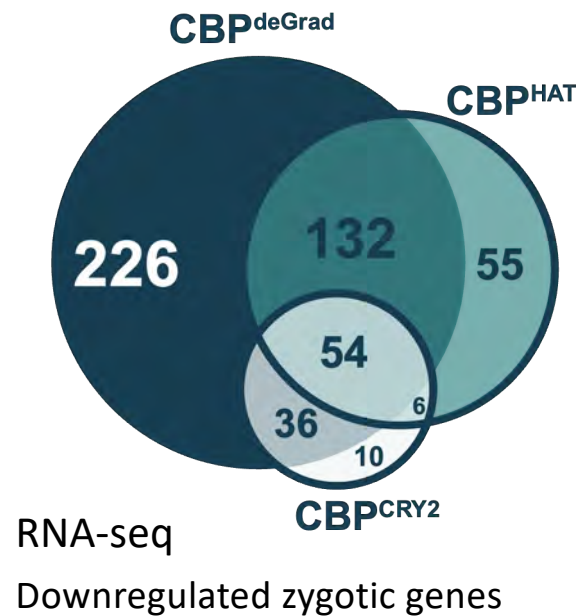
CUT&Tag



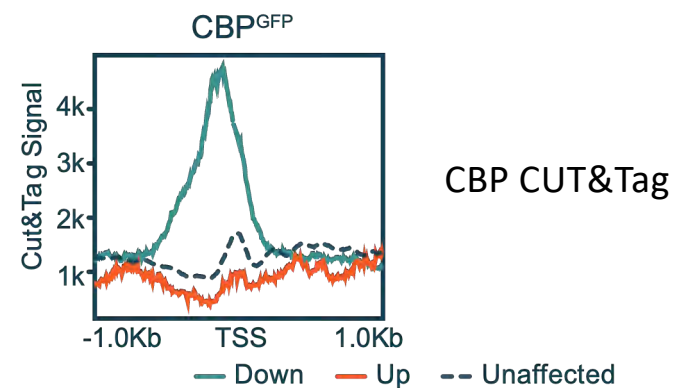
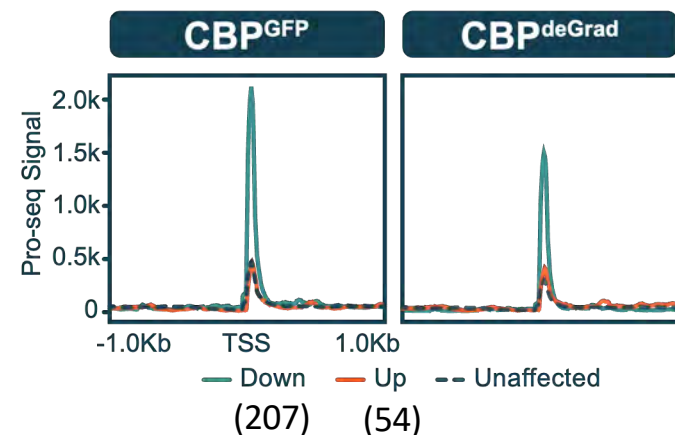
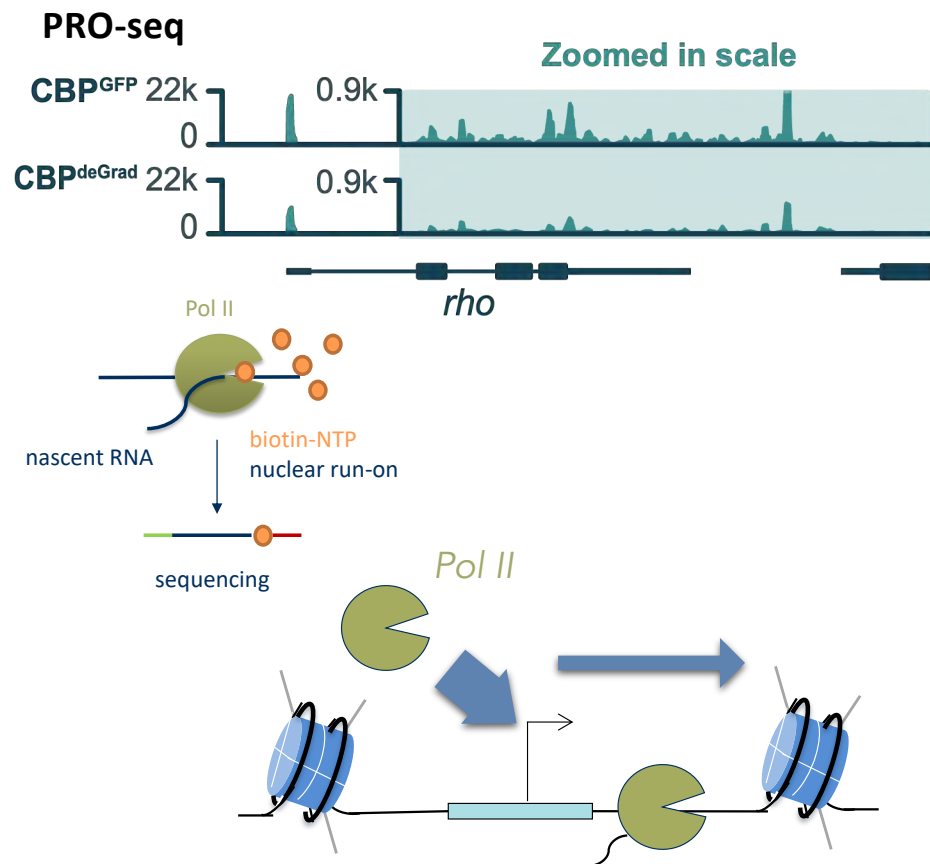
Both mutants still bind chromatin, but block embryogenesis, demonstrating that CBP catalytic activity is required for embryo development



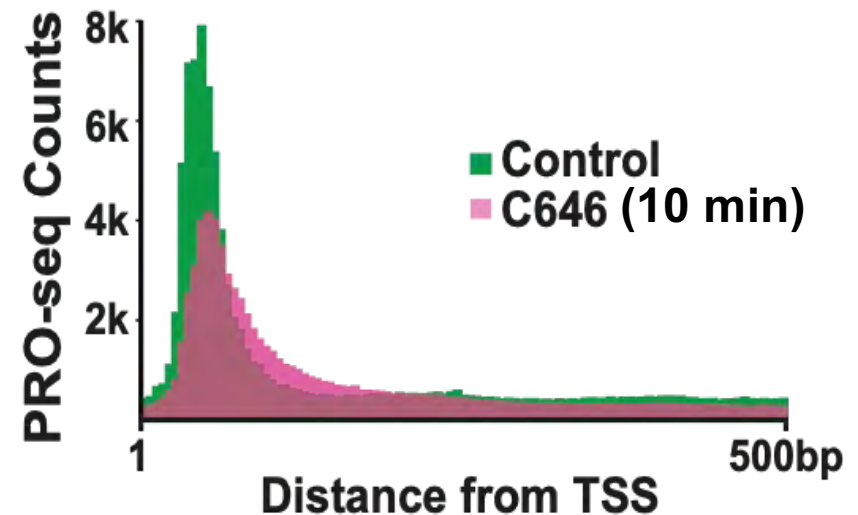
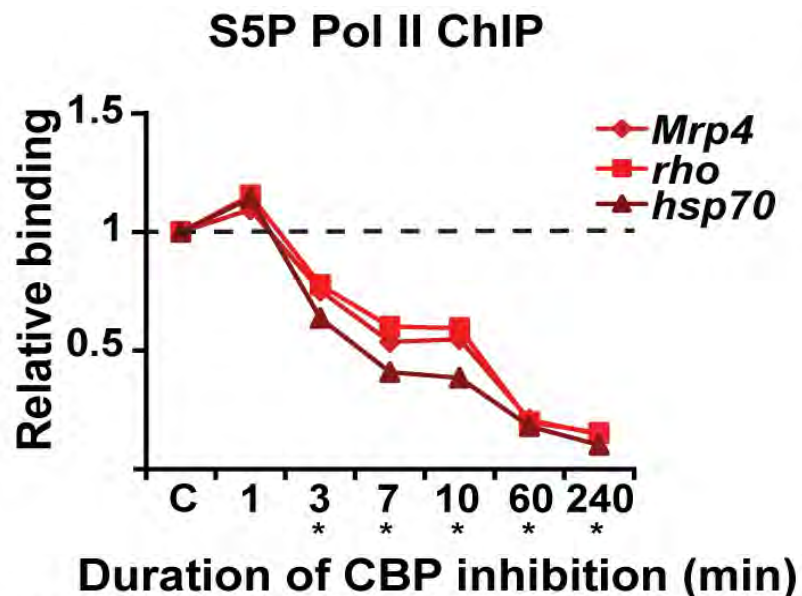
Catalytic-dependent and independent zygotic genes



Precision run-on sequencing (PRO-seq) shows that CBP-regulated genes are highly paused



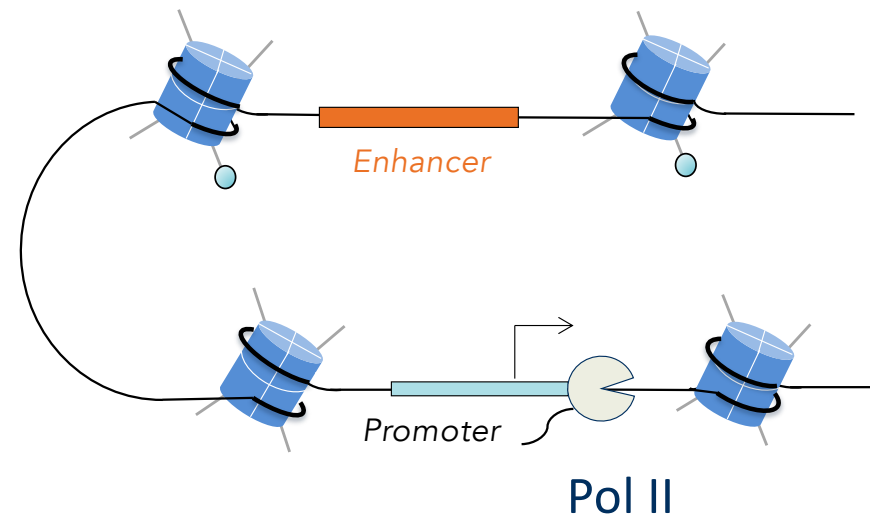
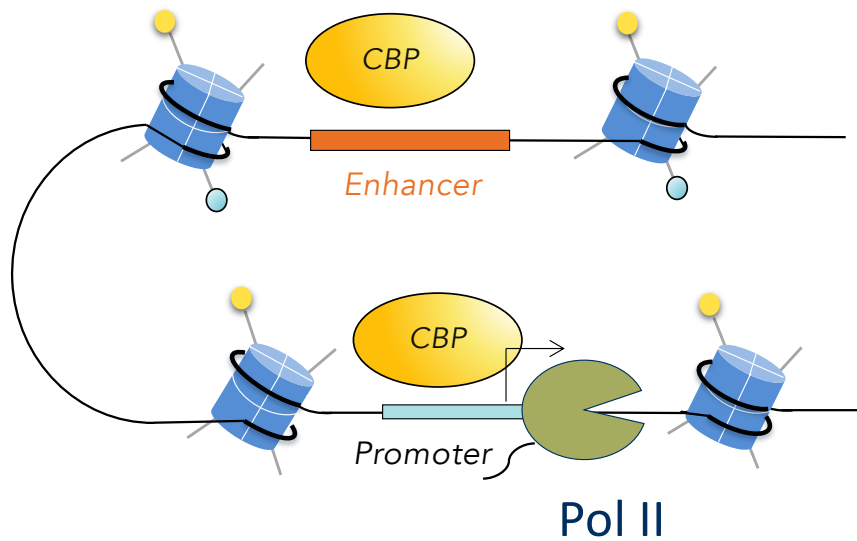
CBP inhibition with the chemical C646 leads to reduced Pol II promoter occupancy



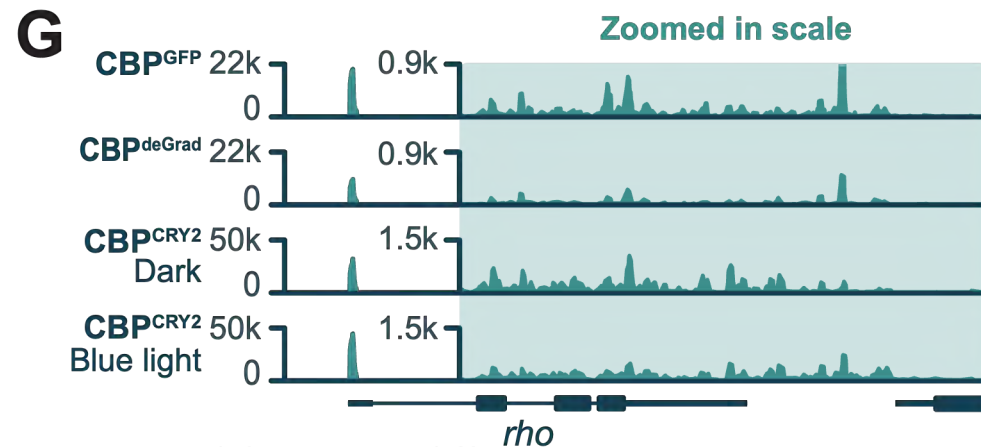
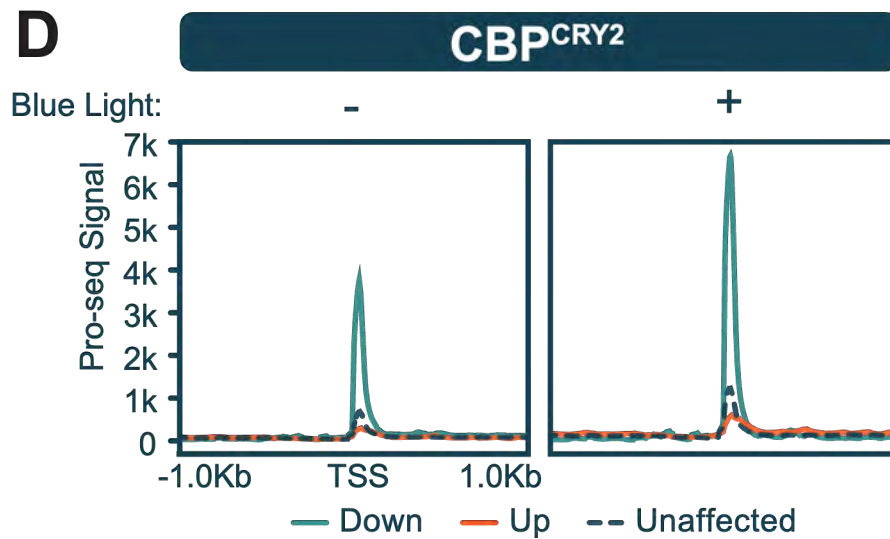
S2 cells



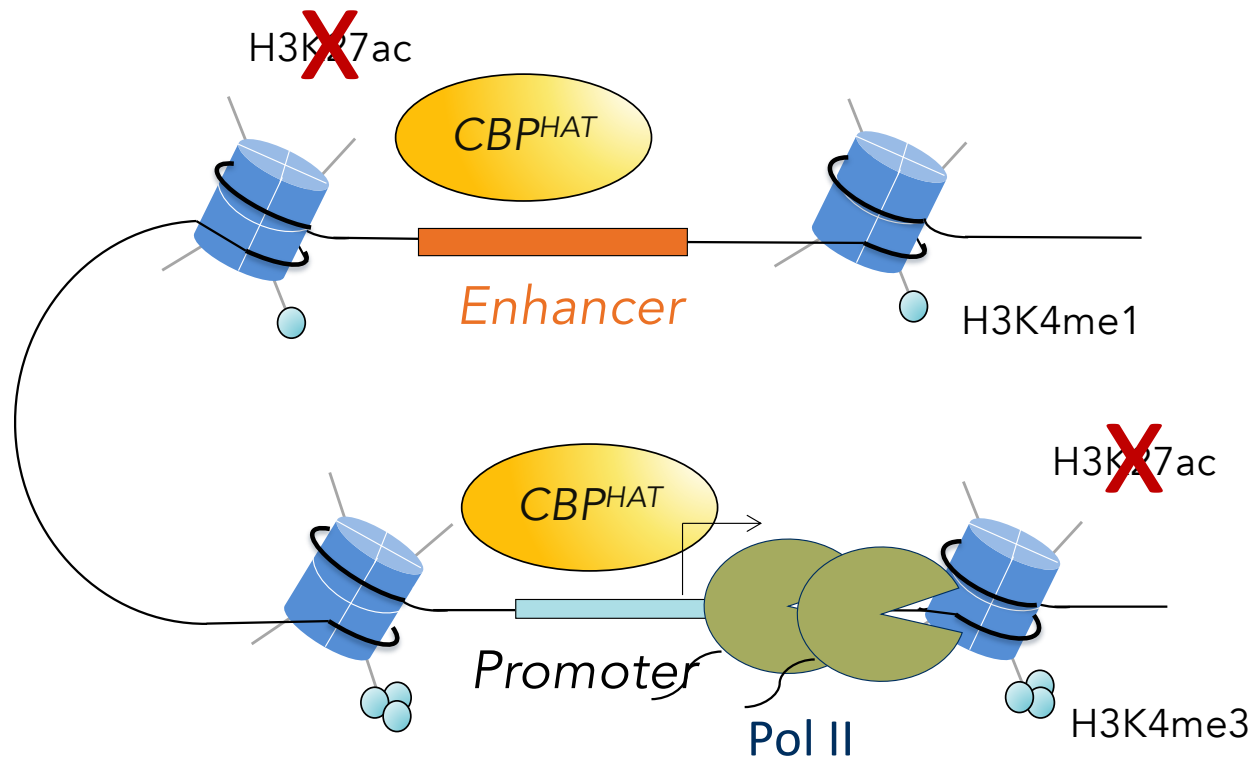
CBP stimulates Pol II recruitment/occupancy



CBP catalytic activity is needed for pause release



CBP catalytic activity stimulates Pol II pause release



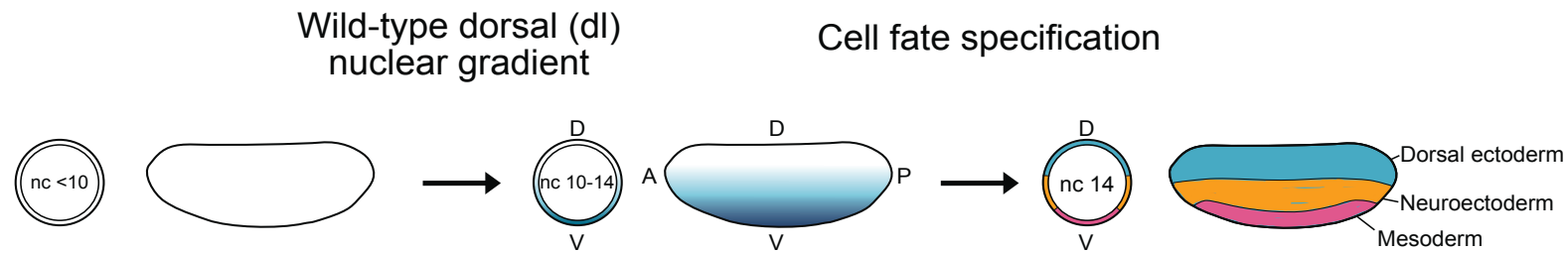
Summary I

- The pioneer factor Zelda is necessary for occupancy of cell fate determining transcription factors
- Zelda is also necessary for CBP to occupy a subset of genomic binding sites
- Neither CBP nor the histone acetylations that it catalyze are required for Zelda to make chromatin accessible
- CBP is important for zygotic genome activation
- CBP recruits RNA Pol II to promoters through non-enzymatic means
- CBP catalytic activity is needed for efficient Pol II pause release and embryo development

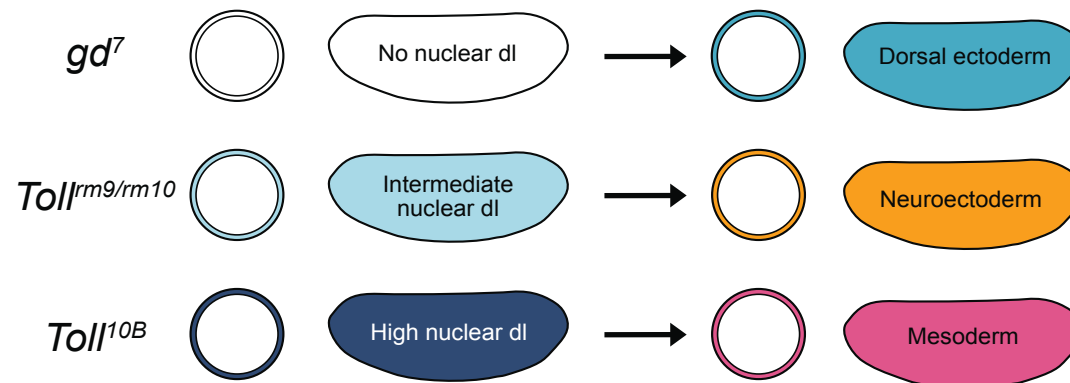
What happens in different cell types?



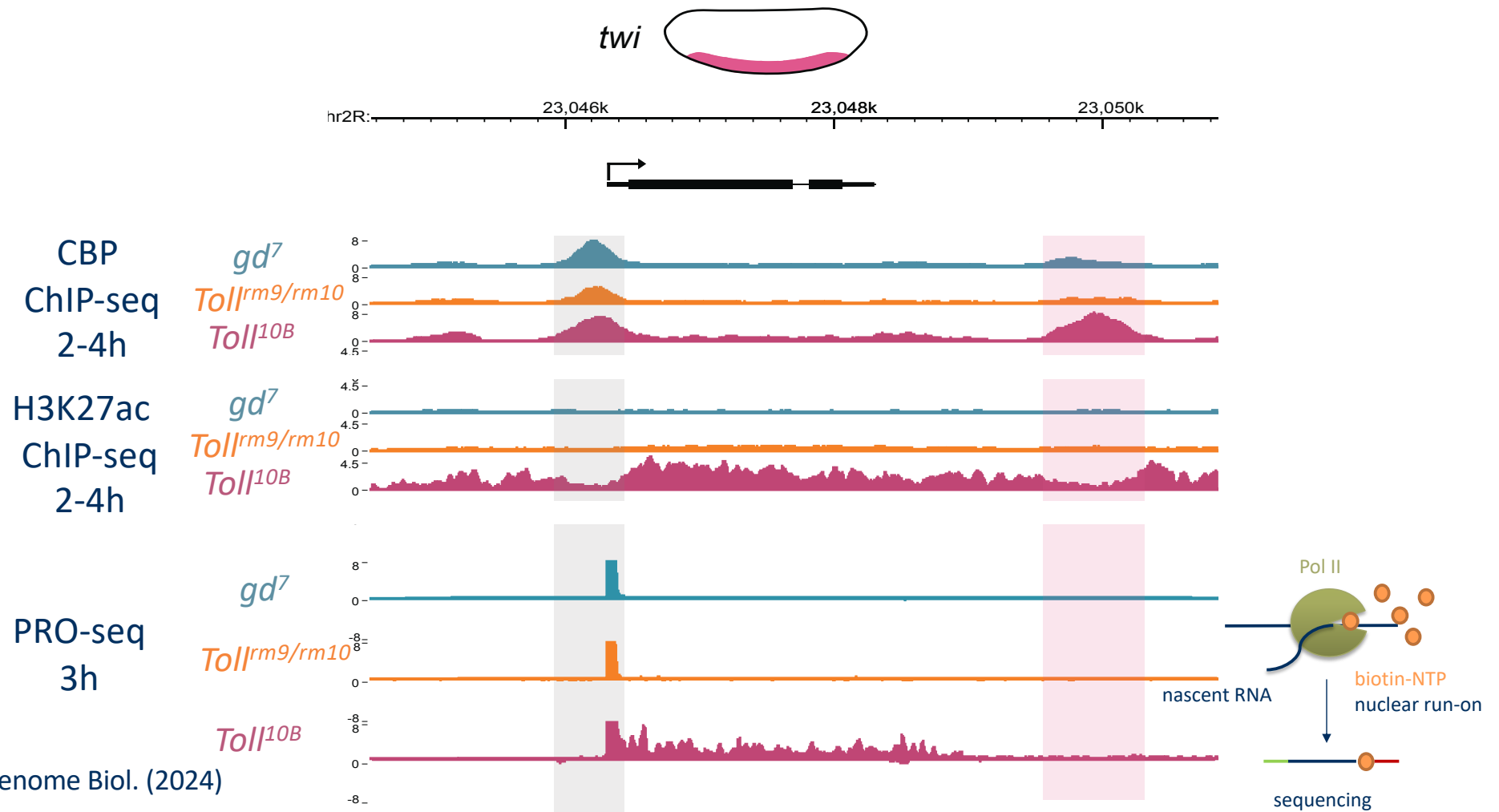
Dorso-ventral patterning of the *Drosophila* embryo



Toll signaling mutants



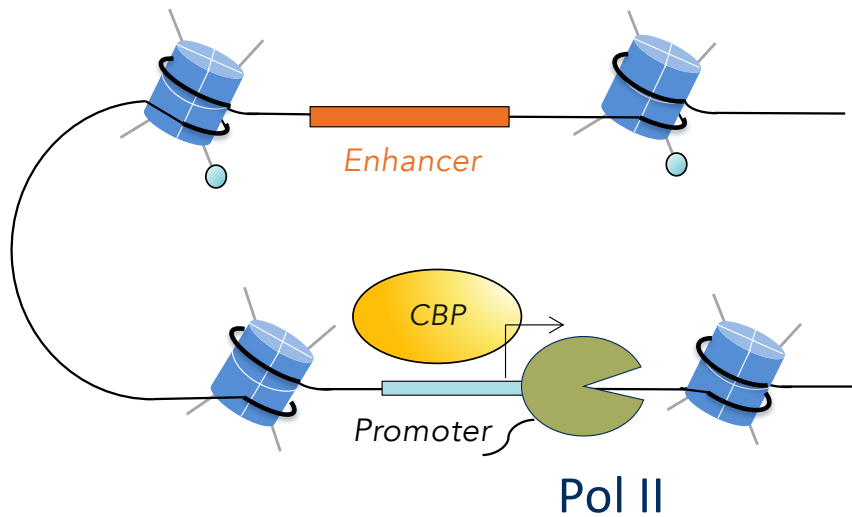
CBP binds enhancers tissue-specifically, but occupies promoters tissue-invariantly



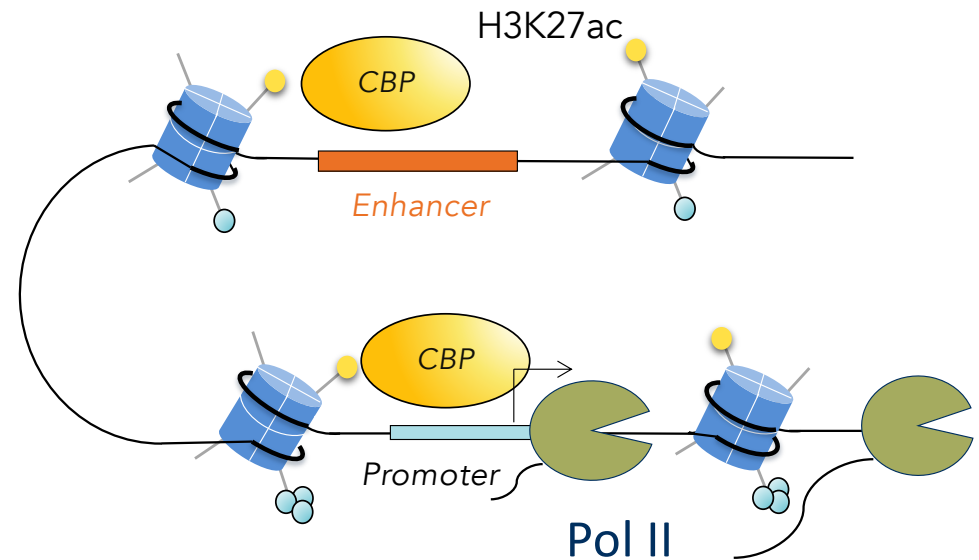
Hunt et al. Genome Biol. (2024)

CBP occupies enhancers tissue-specifically and stimulates Pol II promoter occupancy tissue invariantly

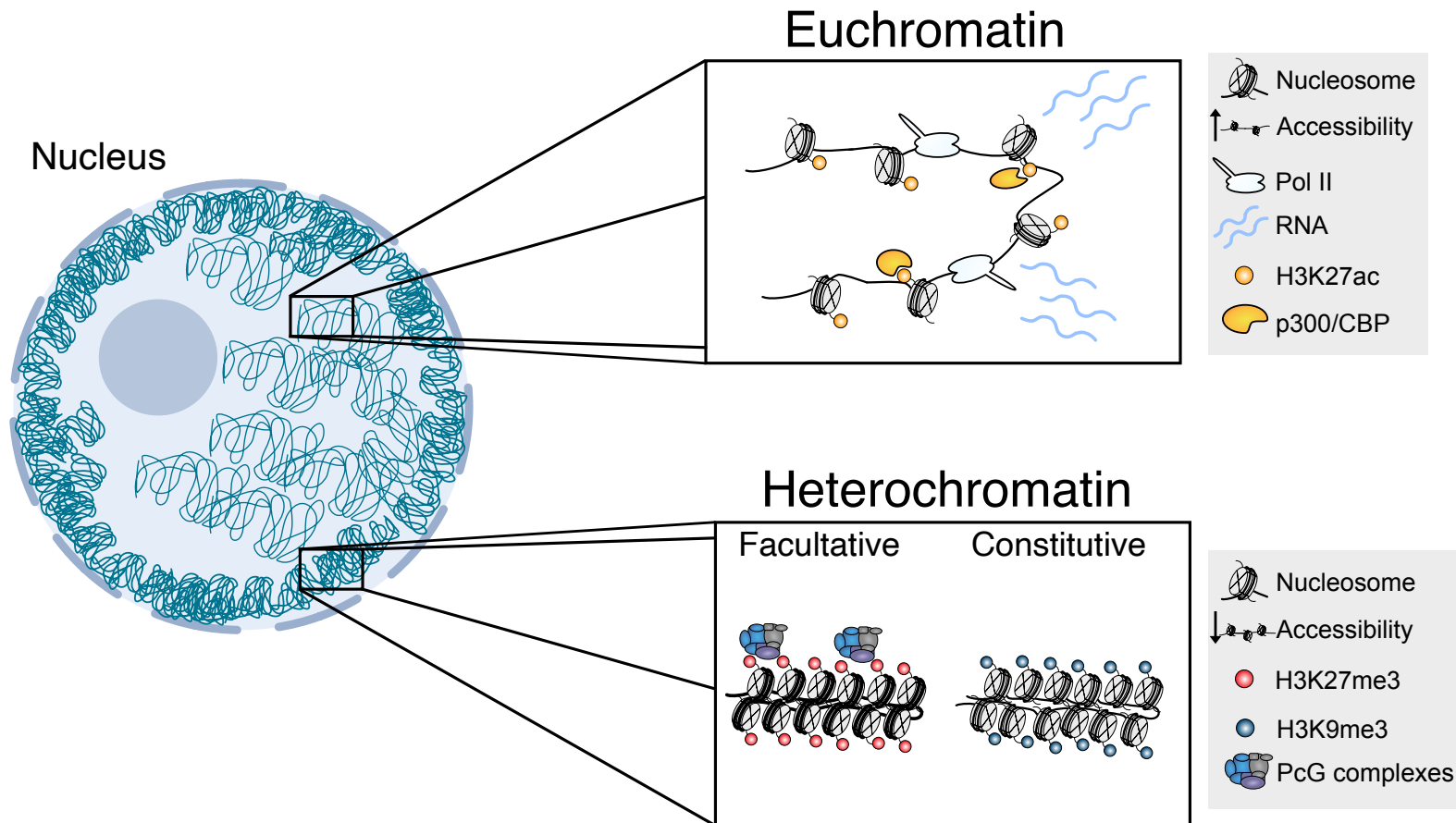
**Developmental gene in
silent cell type**



**Developmental gene
in active cell type**

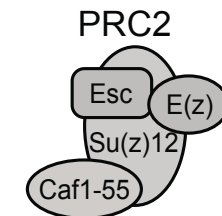
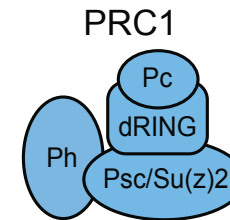


H3K27ac is present at active genes, whereas H3K27me3 is used to silence genes

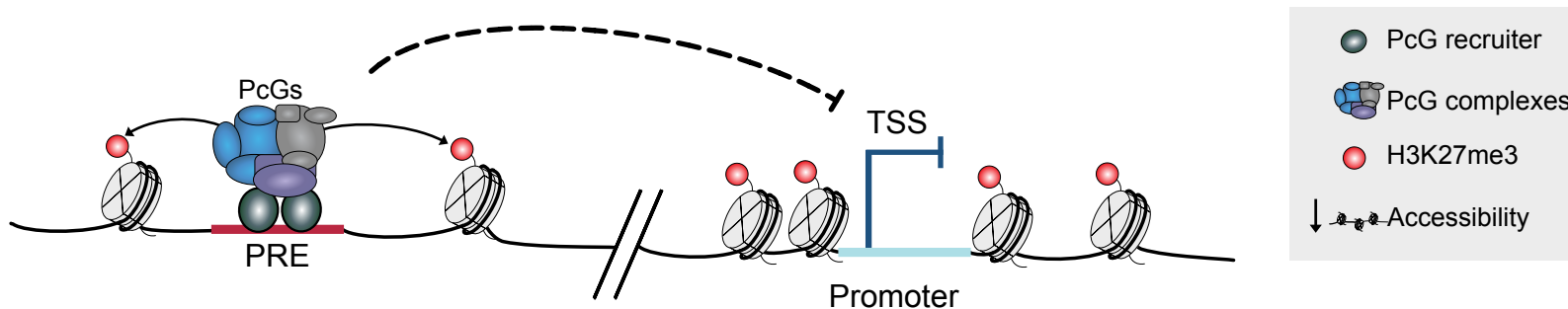


Polycomb group (PcG) proteins silence transcription of genes

-Highly conserved proteins that form multi-subunit complexes to maintain repression of developmental genes



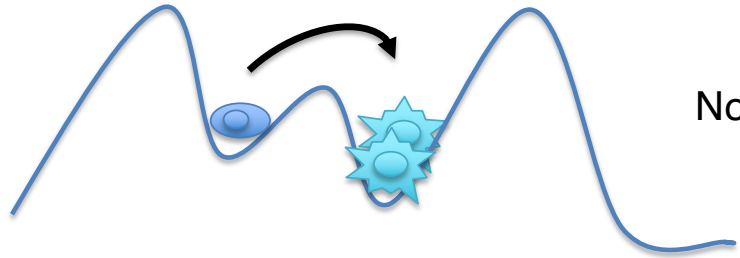
Polycomb mediated repression



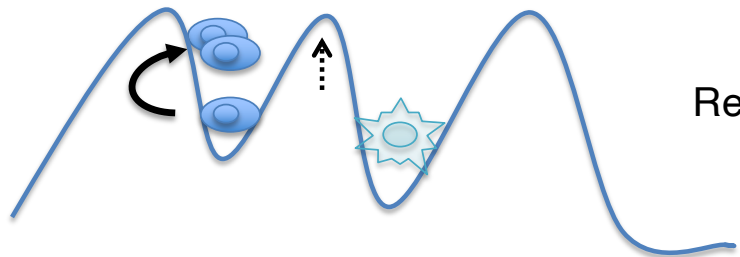
PRC2 catalyzed H3K27me3 is a hallmark of repressive facultative heterochromatin

Cell state transitions

Normal

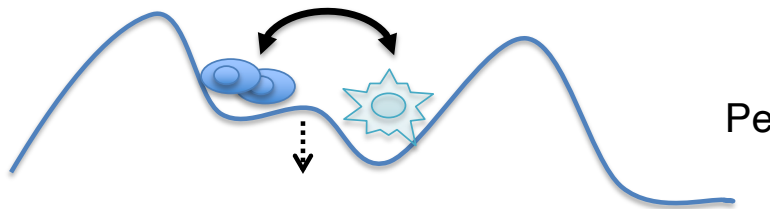


Restrictive



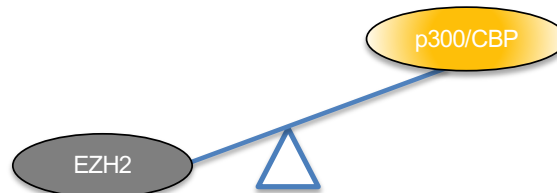
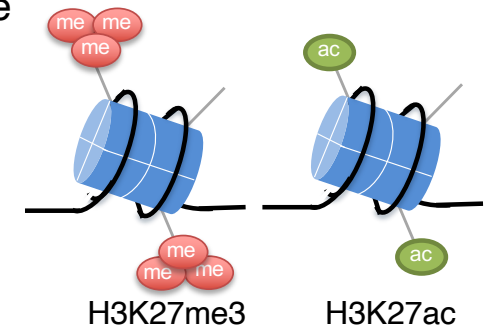
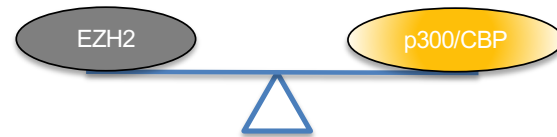
Raised chromatin barrier
Cell unable to leave proliferative state

Permissive

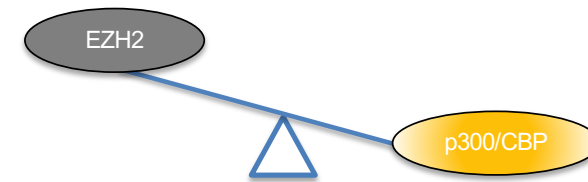
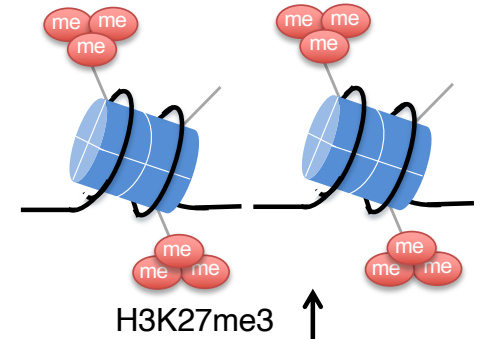


Lowered chromatin barrier
Epigenetic plasticity allows bidirectional transitions

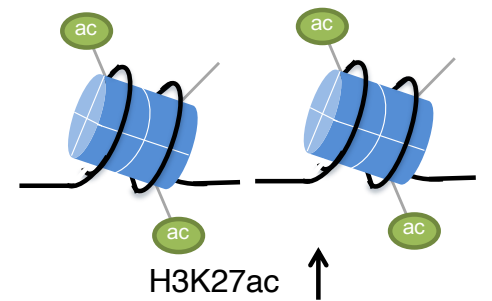
Epigenetic state



EZH2 GOF mutation or p300/CBP LOF mutation



EZH2 LOF mutation or p300/CBP upregulation

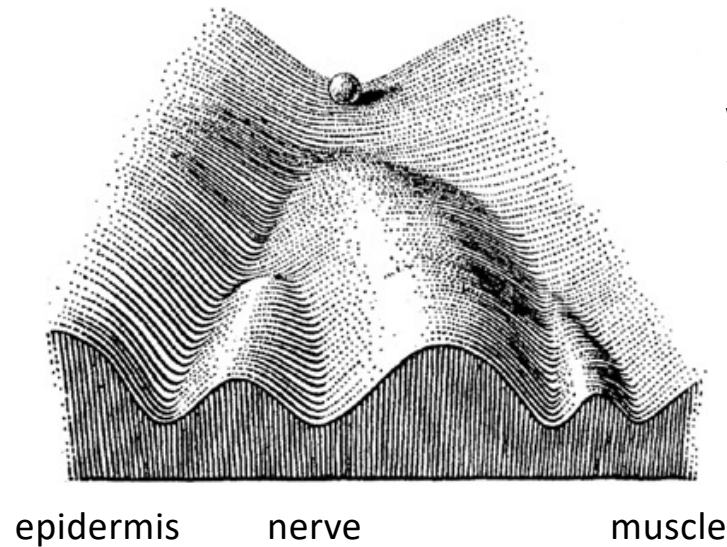


Conrad Hal Waddington

1905-1975



Epigenetic landscape



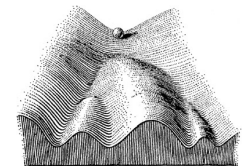
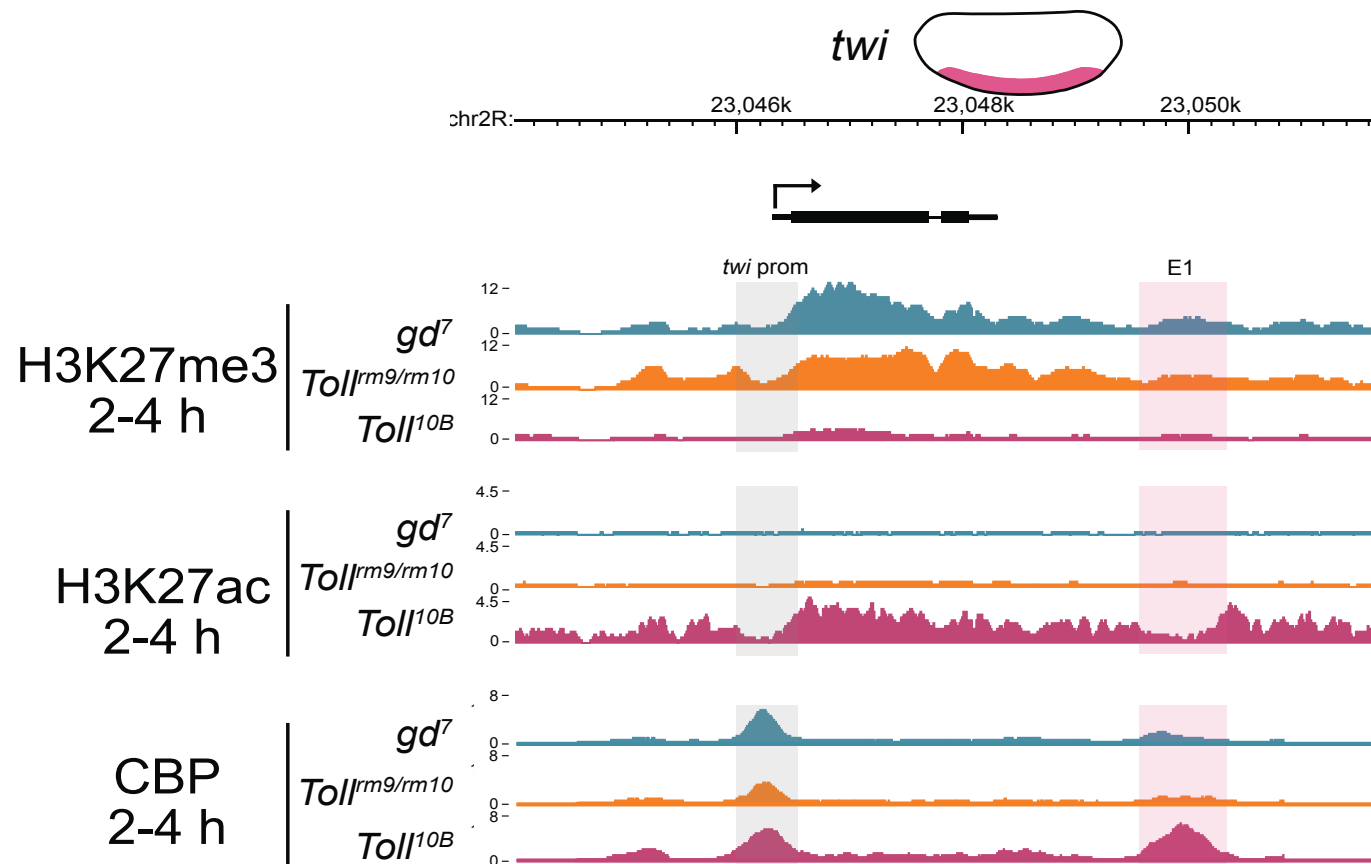
Waddington, C.H. 1968. The basic ideas of biology.

Waddington, C.H. 1957. The strategy of the genes.

epigenesis + genetics = epigenetics

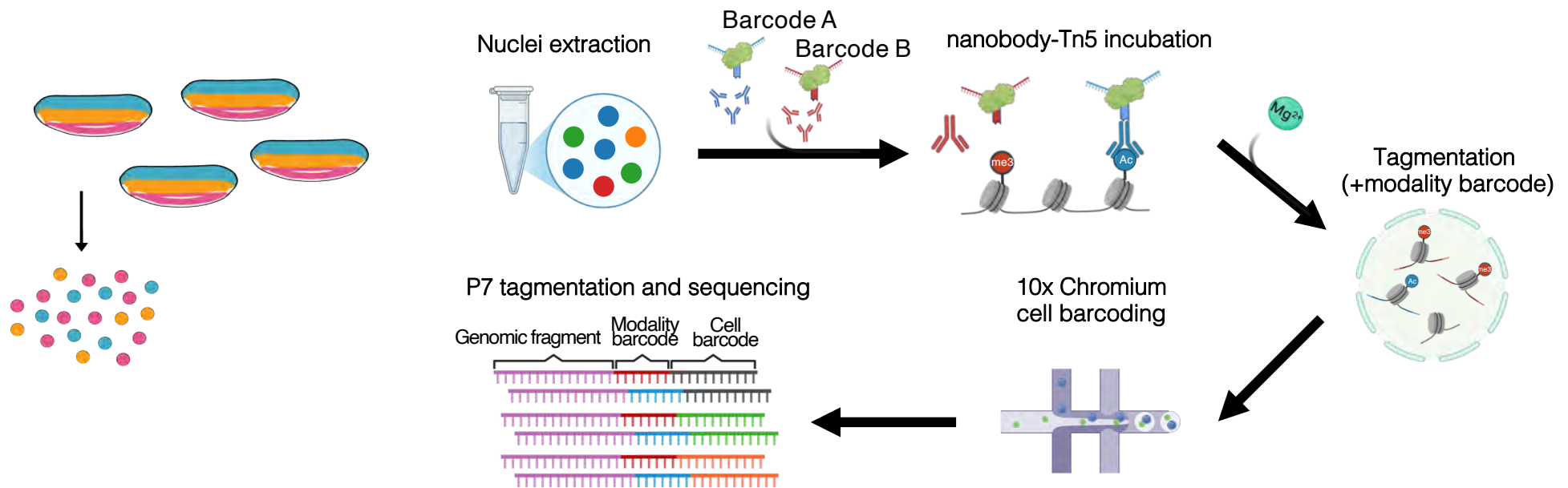


H3K27me3 in silent tissues, H3K27ac in active tissue



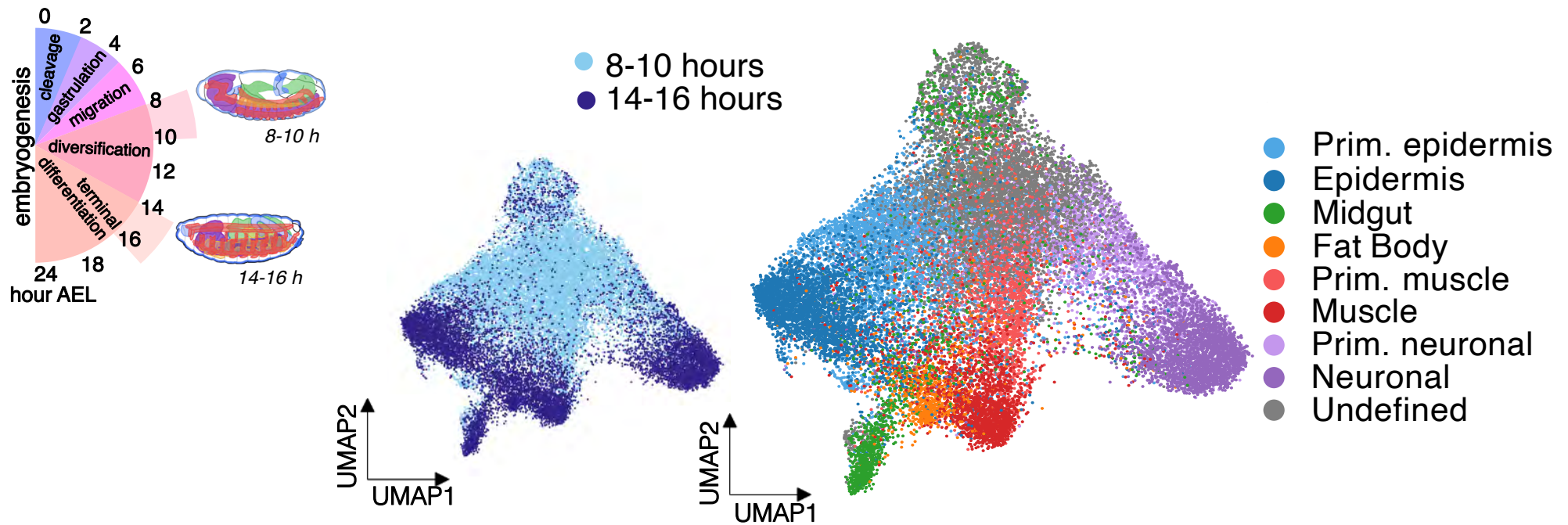
Hunt et al. Genome Biol. (2024)

Nano CUT&Tag – single cell chromatin profiling



Nano CUT&Tag can separate stages and cell types from histone modification data

UMAP based on H3K27ac and H3K27me3

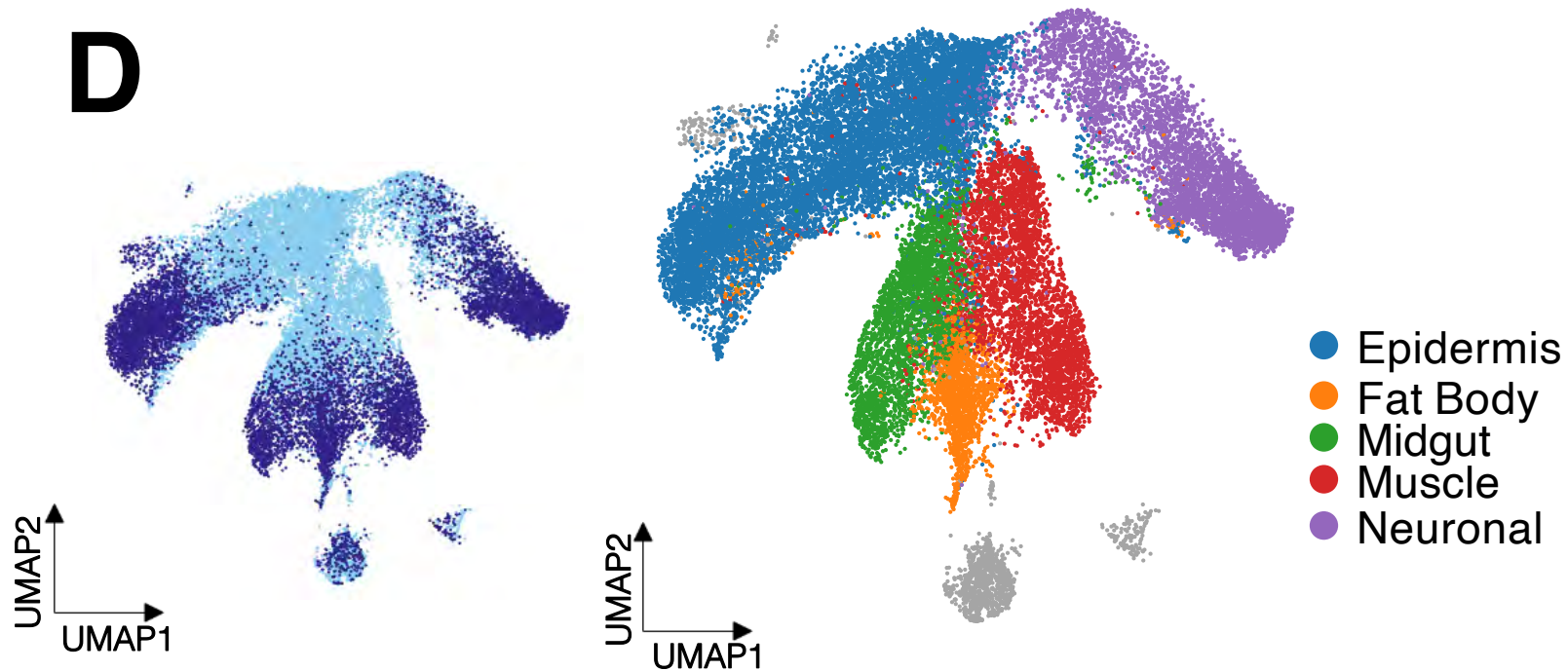


Two stages, two replicates, $\approx 27\,000$ cells

24 000 5kb bins only a few hundred fragments per cell

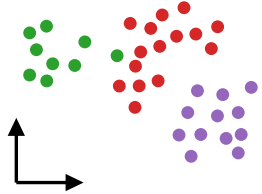
Pirogov et al. bioRxiv

Integration with published scATAC-seq data

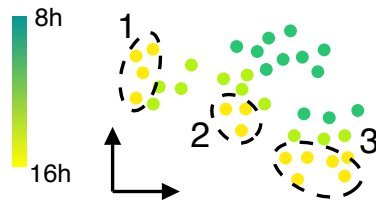


Generation of a chromatin landscape

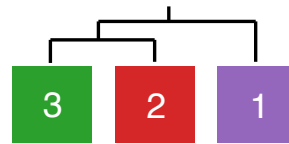
2D UMAP, cells colored based on cell-type



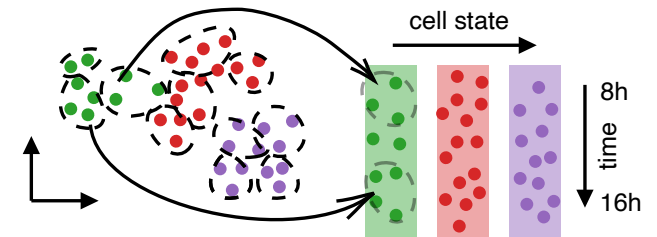
1. Determine cell types at the late stage



2. Create hierarchical cell type scaffold



3. Assign cells to scaffold by their resemblance to cell types and according to their age

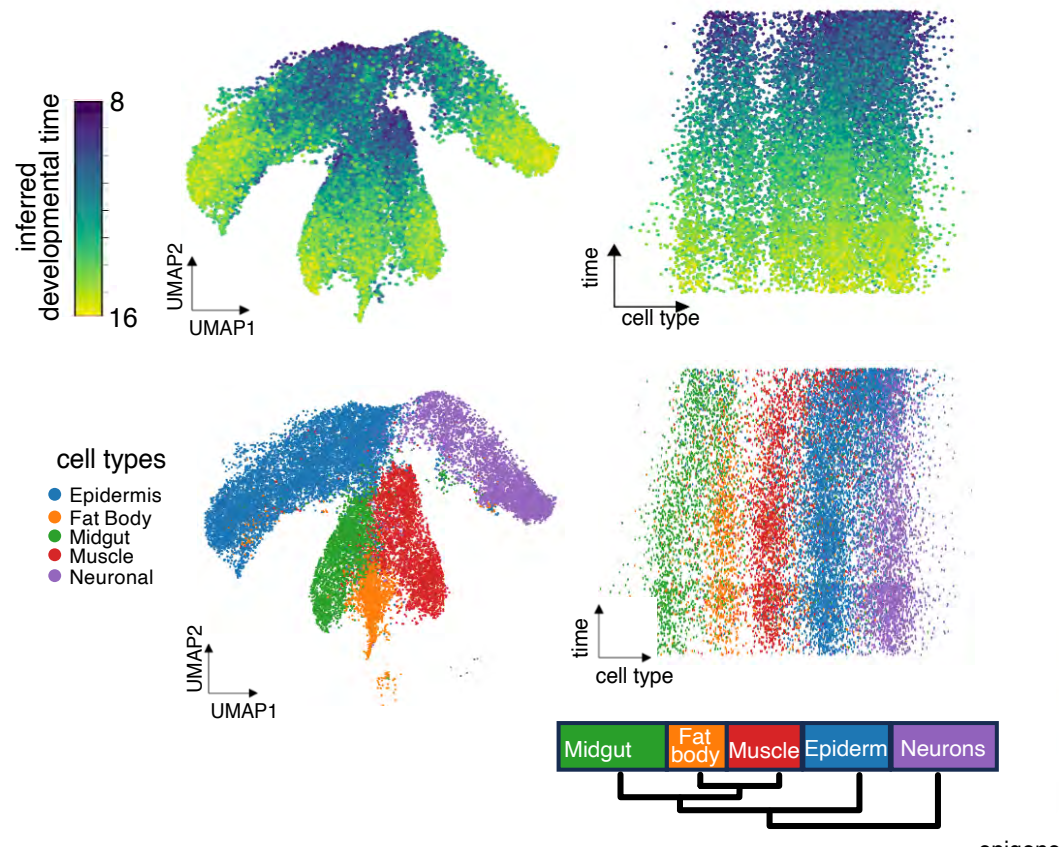


Sergei Pirogov



Aleksander Purik

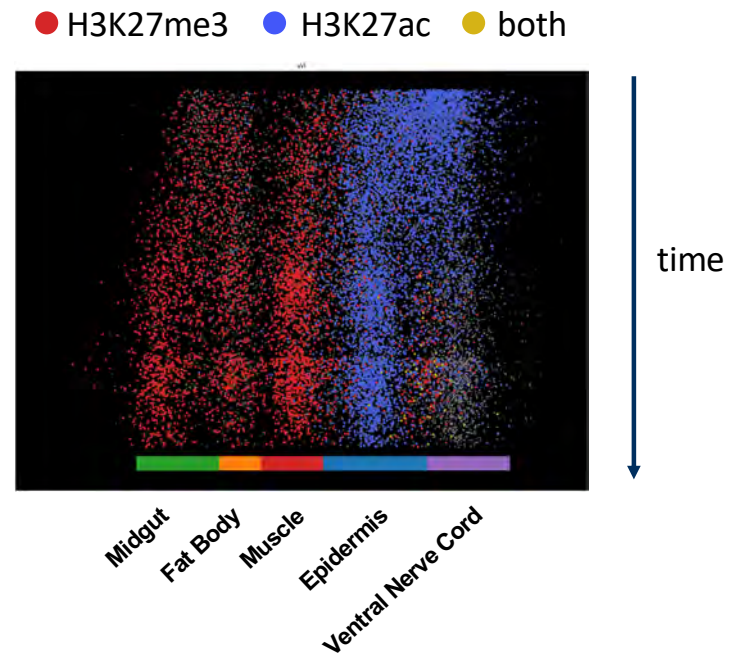
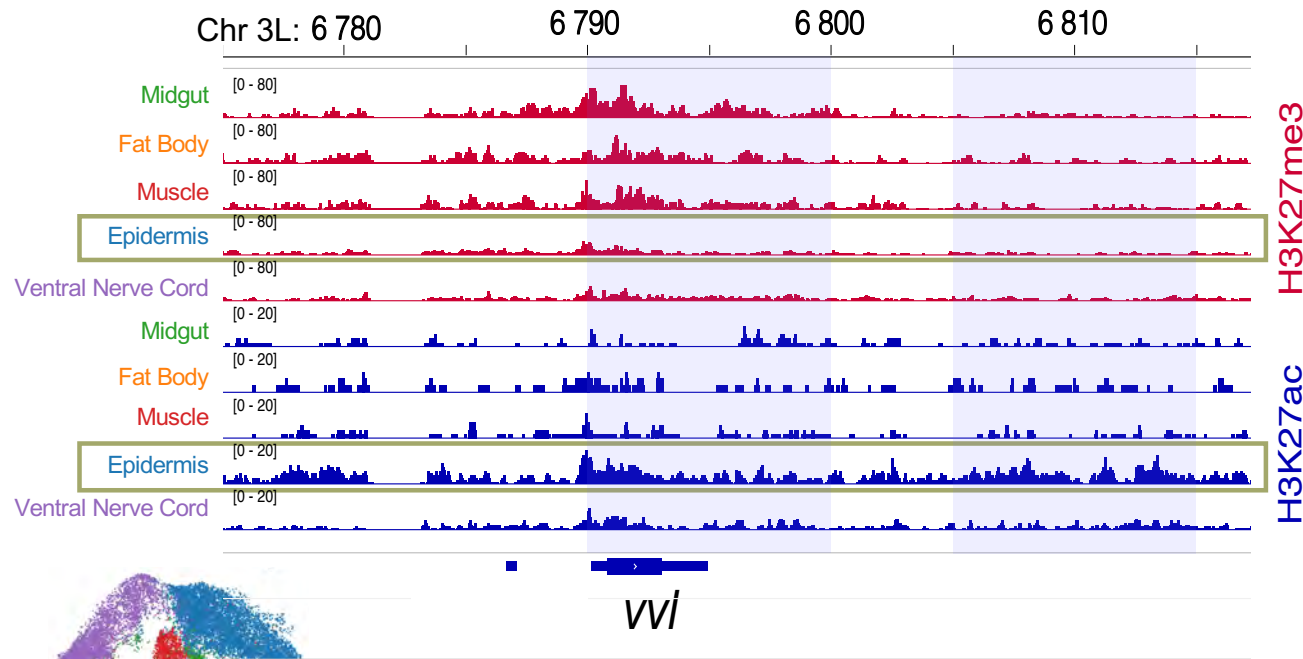
Generation of a chromatin landscape



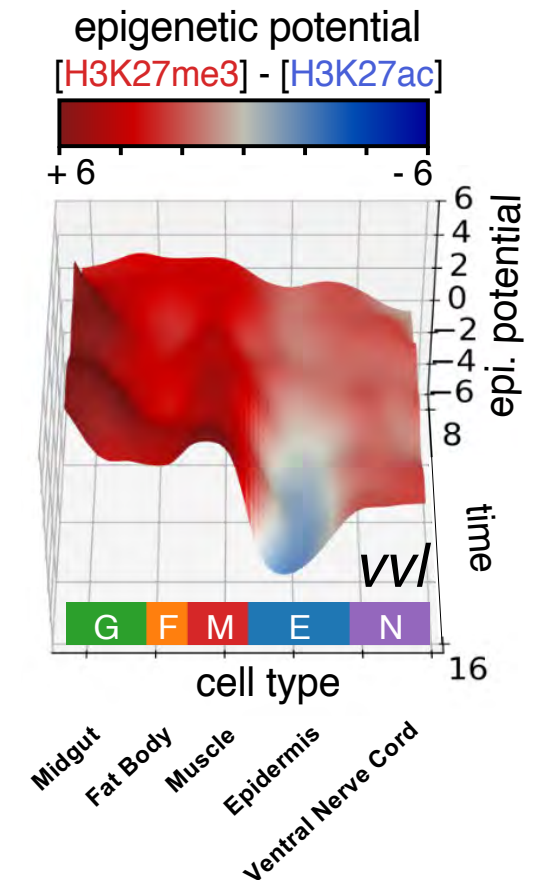
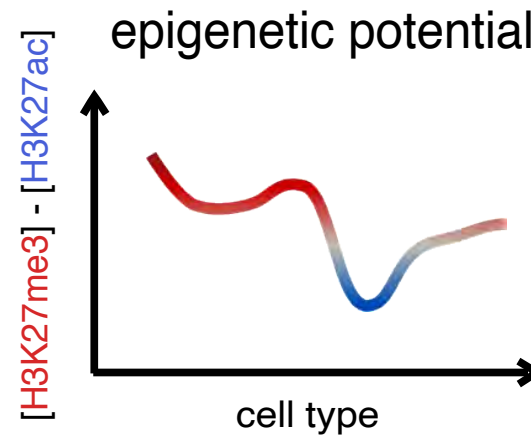
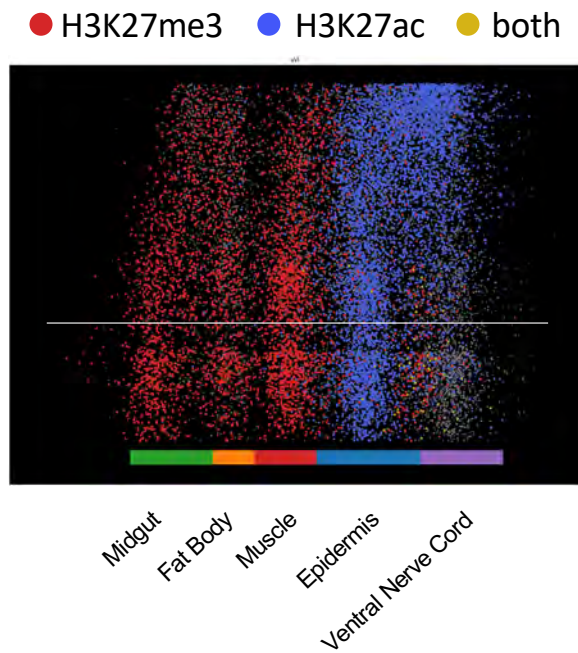
Chromatin landscape of *ventral veins lacking (vvl)* gene

Required for trachea formation and patterning of the ventral ectoderm

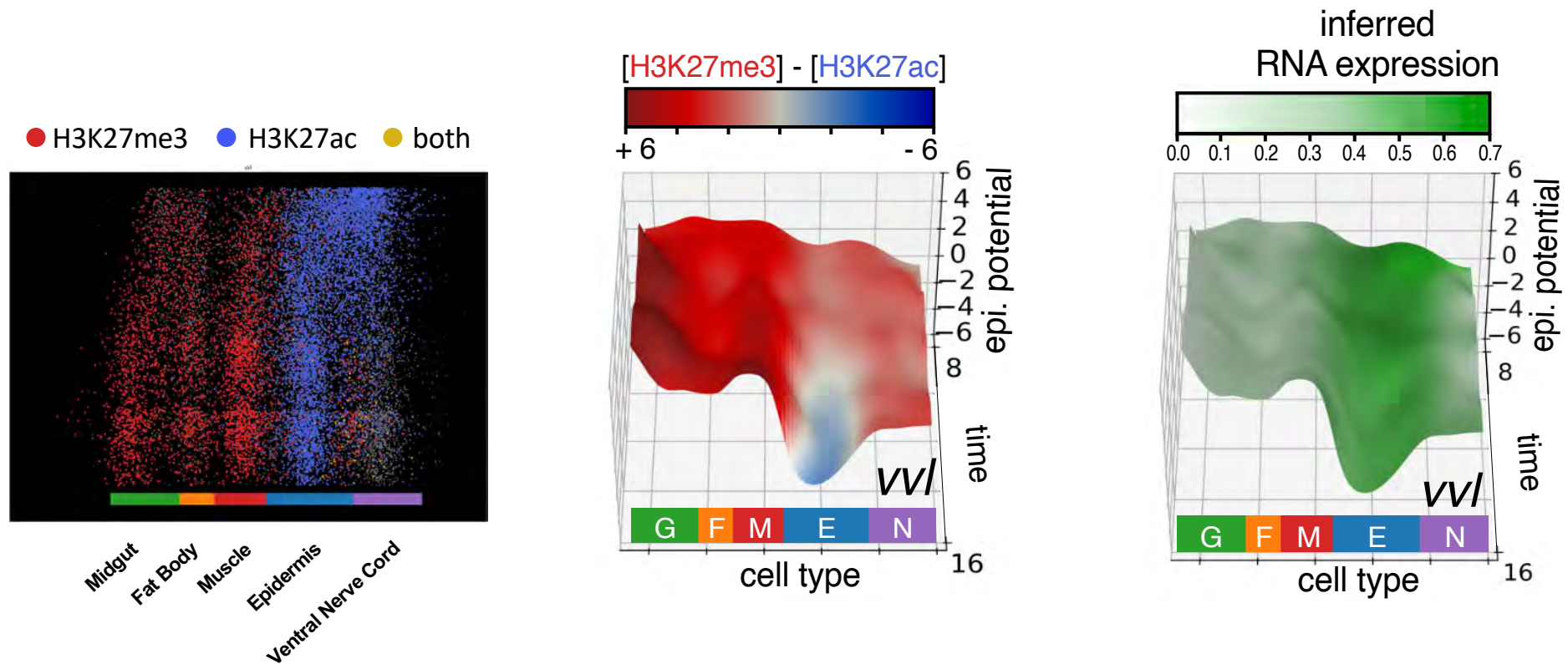
14-16h



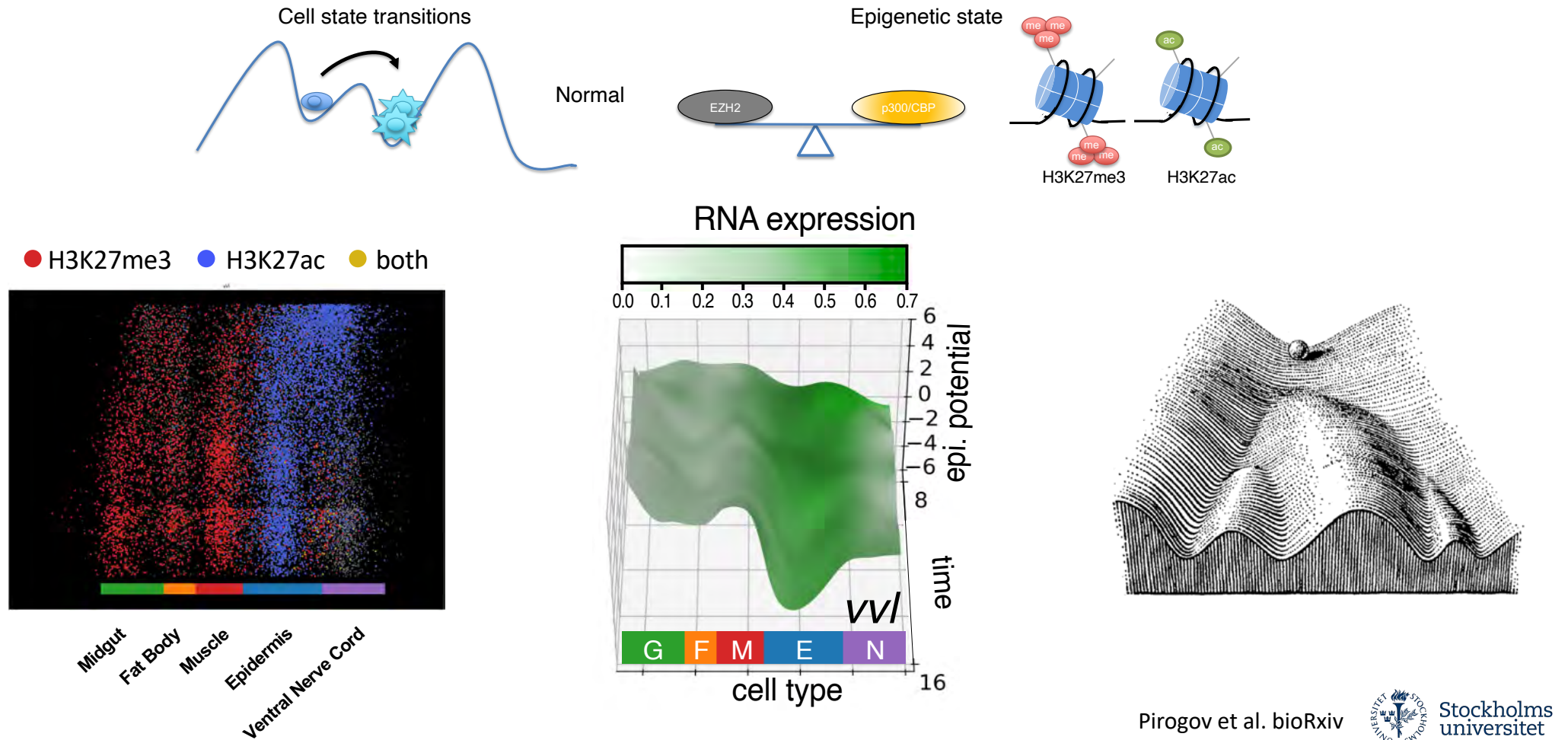
Chromatin landscape of *ventral veins lacking (vvl)* gene



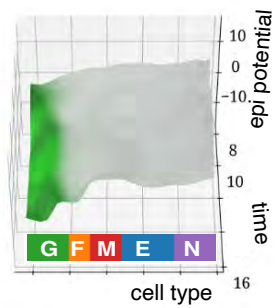
Chromatin landscape of *ventral veins lacking (vvl)* gene



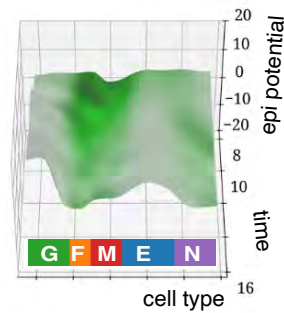
Chromatin landscape of *ventral veins lacking (vvl)* gene



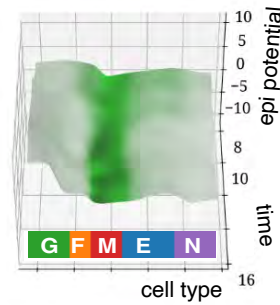
More chromatin landscapes



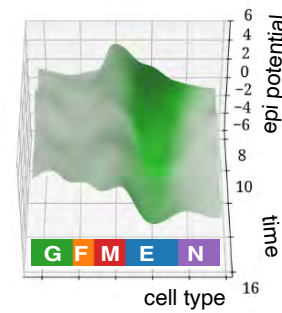
GATAe



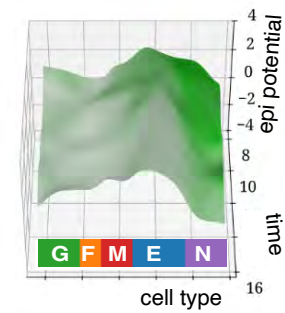
zfh1



vn

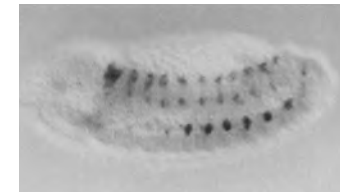
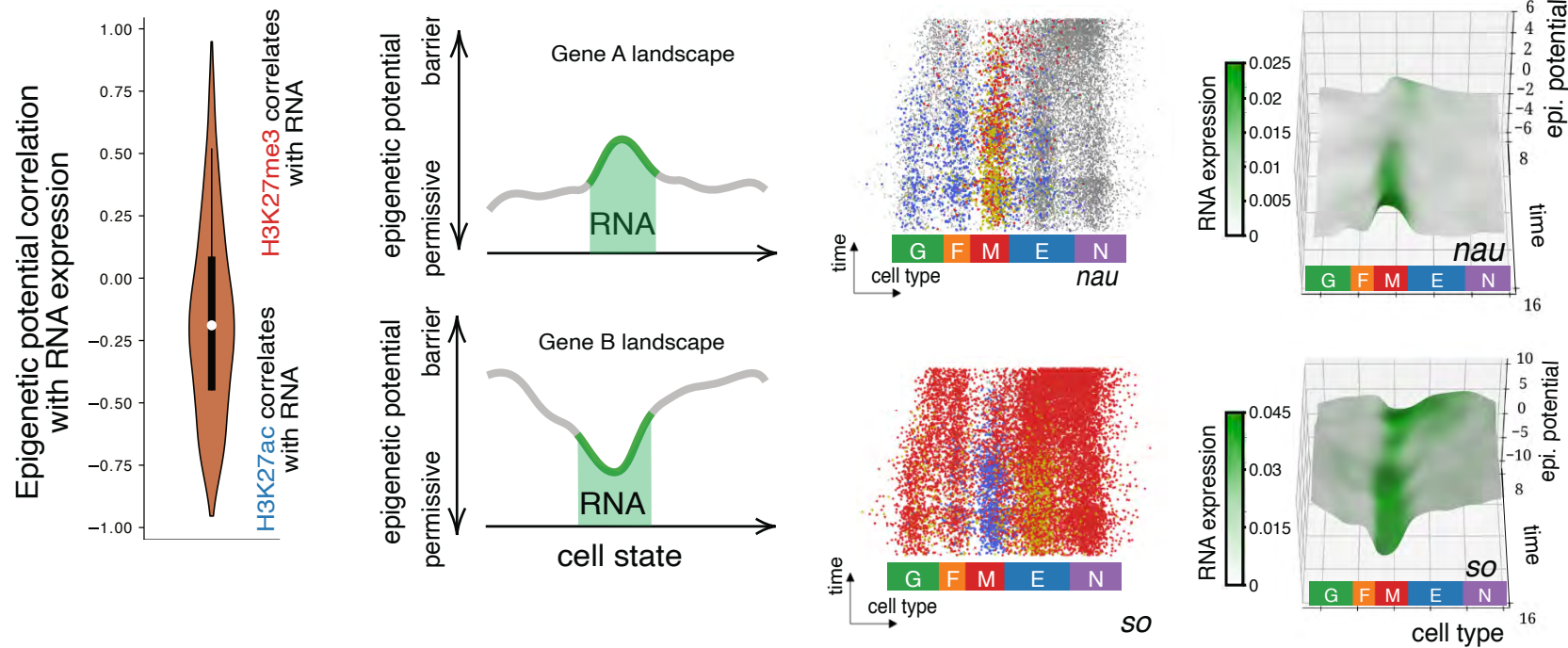


grh

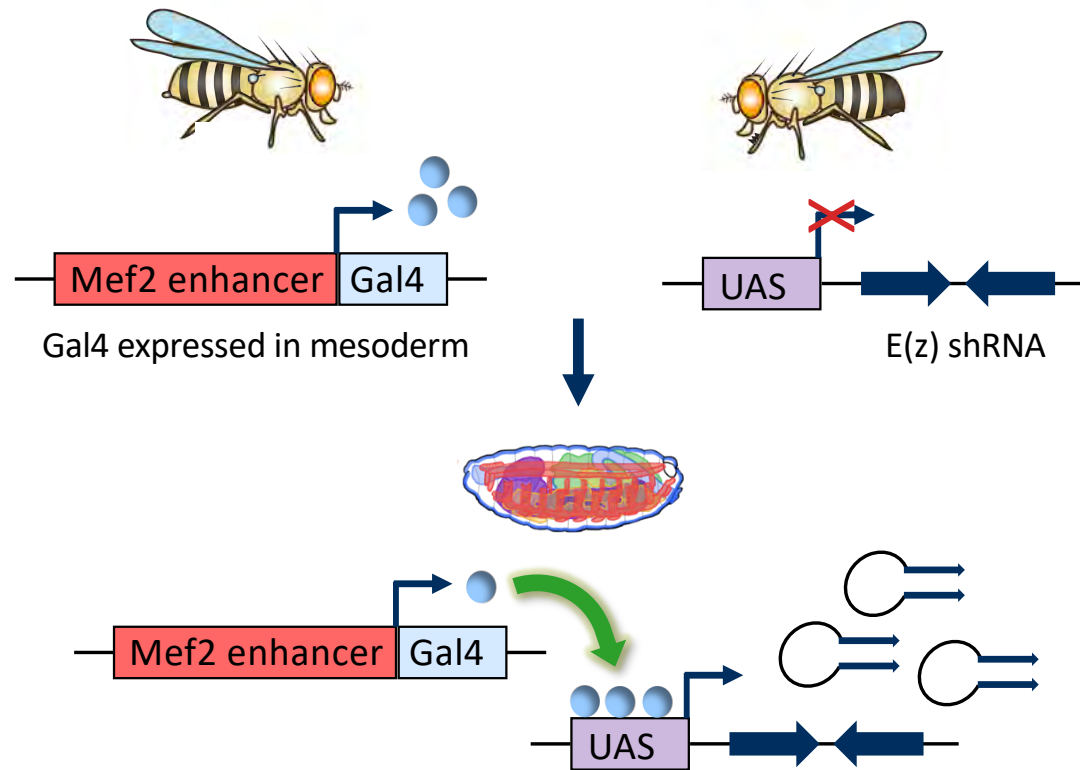


Ptp61F

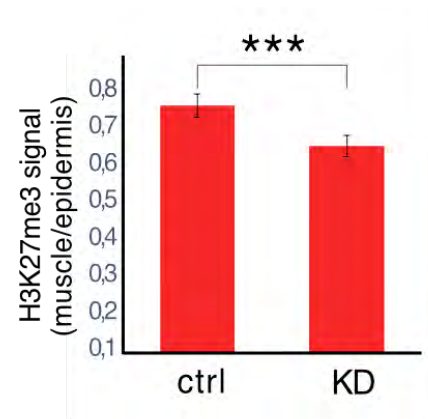
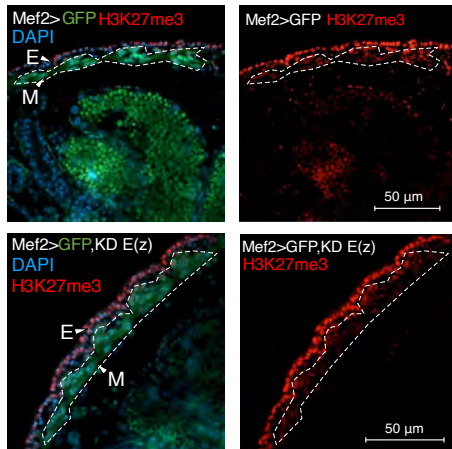
Co-occurrence of H3K27me3, H3K27ac and RNA expression in the same cell lineage at many genes



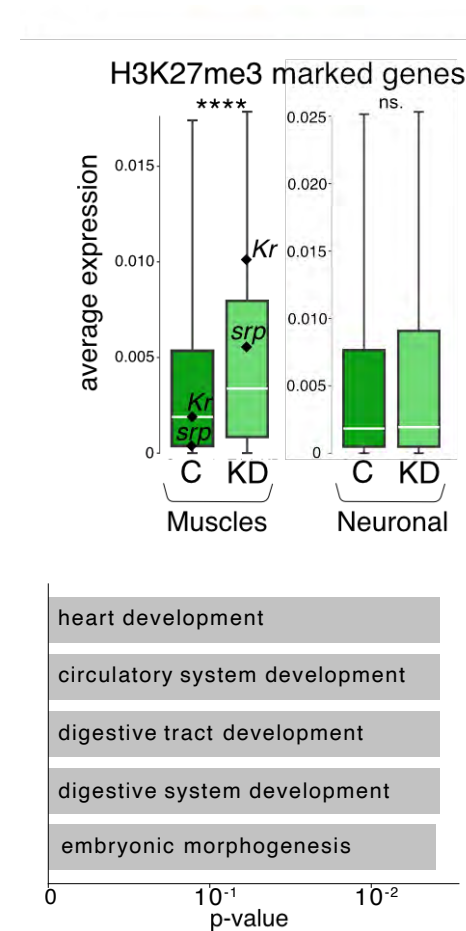
Knock-down of E(z) in mesoderm using Mef2-Gal4



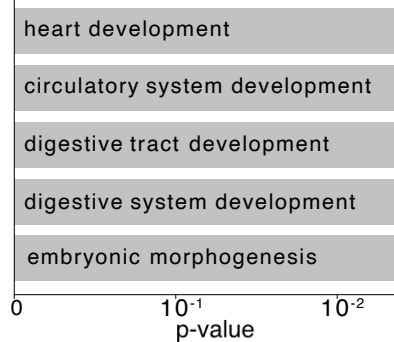
Single nucleus RNA-seq shows that many up-regulated genes in Mef2>E(z) KD mesoderm are normally expressed in the mesoderm lineage



H3K27me3 immunostaining



snRNA-seq



Chromatin barriers within a tissue is common for differentiated cells

Naïve embryo



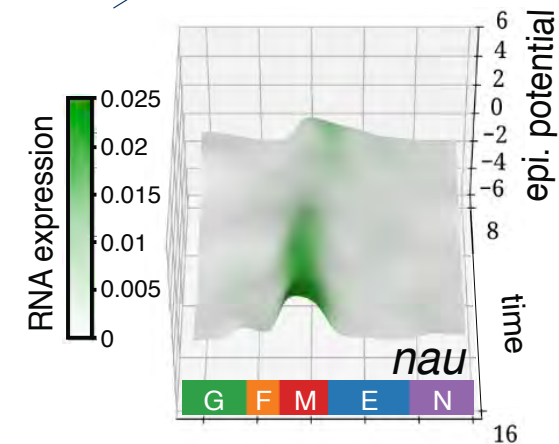
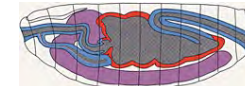
Specified



Determined



Differentiated



Chromatin barriers between cell types may be common during cell specification

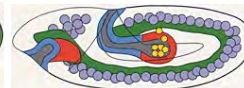
Naïve embryo



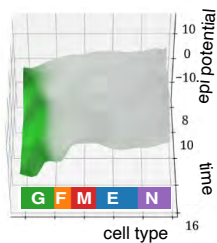
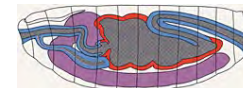
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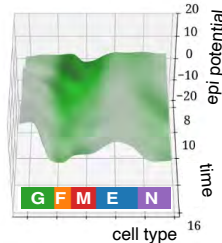
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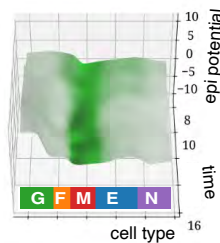
Differentiated



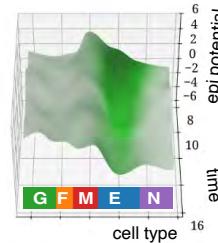
GATAe



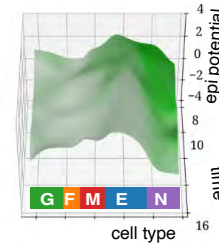
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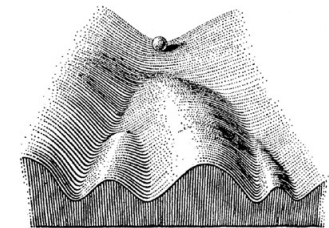
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grh



Ptp61F



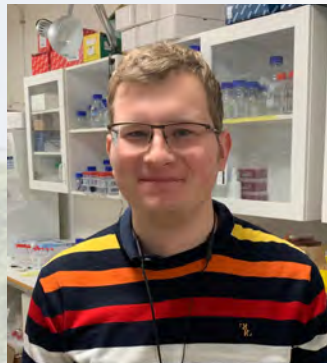
Summary II

- The balance between CBP-mediated H3K27ac and PRC2-mediated H3K27me3 may contribute to the formation of chromatin landscapes
- Chromatin landscapes can be identified by simultaneously mapping H3K27ac and H3K27me3 in single cells by nano-CUT&Tag
- Chromatin barriers may restrict gene expression to specific cell types
- H3K27me3 is commonly found in the same cell lineage as the gene that is expressed and may be involved in preventing expression in other cell-types in the lineage

Acknowledgements



George Hunt



Sergei Pirogov



Aleksander Purik



Artem Ilin



Melissa Harrison
Audrey Marsh
University of Wisconsin - Madison

Marek Bartosovic
Jose Ramon Barcenas Walls
Stockholm University - DBB



Cancerfonden 